

claude-windows / accd47af-95a3-46a7-9b51-b3eb5066a777

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\subagents\workflows\wf_793a5d86-1c4\agent-a5db6aa983d198745.jsonl`  
- SHA-256: `2ce5a77b1a63042a7c430d99dd5ddfa487b032db32120f121c3e5a3aa93981cb`  
- Source modified: `2026-07-02T08:19:35+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_793a5d86-1c4`  
- session_id: `accd47af-95a3-46a7-9b51-b3eb5066a777`
```

Transcript

1. user (2026-07-02T08:16:24.739Z)

You are reviewing OPEN GitHub PR #420 on Khelsutra/badminton-highlight-indexer to decide its fate: should we TAKE IT OVER (adopt/rebase/verify/merge) or CLOSE it as superseded by already-merged work?

PR #420: "docs: reconcile stale trackers/blurbs (R1-13/R1-14/R1-01-doc/DATA_IN_GCS)" (head branch: origin/docs/reconcile-trackers-2026-07-02, mergeable=CONFLICTING)
Investigation hint: Docs-only across DOC_STATUS.md, docs README, NEXT_STEPS, BACKLOG, DATA_IN_GCS, yolo_hybrid docstring. Handoff says #423 already registered plan + reconciled some trackers. CHECK: for each file #420 touches, does master already reflect the same reconciliation (via #423 or others)? Which hunks are still-unapplied vs already-done vs now-conflicting-with-a-different-edit? Decide fully-superseded vs partially-still-valuable.

Already MERGED onto master (verify against actual master code, do not assume):

```
- #422 = fix for --output-dir clobbering config.json output.output_dir (CODE_MAP updated for it)  
- #423 = docs: health-remediation plan added + DOC_STATUS/README/NEXT_STEPS registered; audit marked snapshot  
- #424 = P2: /api/process idempotent by content hash (#340 re-bill fix)  
- #425 = P3: /api/source resolves remote source into output-scoped source_cache, reuses across Range requests, warms annotation proxy  
- #426 = P4: bounded ffprobe/ffmpeg subprocess timeouts on serving paths  
- #427 = P5: clean remote-input scratch dirs after process/compile/assets/CLI; durable reports  
- #428 = P1/P6: backend/storage/gcs.py lazy-import mpy typing fix  
master HEAD = d00b4db "fix gcs storage lazy import typing (#428)"
```

Working dir: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer (currently on master @ d00b4db). This is a SHARED checkout used by sibling sessions ? DO NOT checkout branches, DO NOT modify the working tree, DO NOT run git merge/rebase/add/commit. Read-only investigation only.

Do this:

1. Get the PR's intended change: run `gh pr diff 420 -R Khelsutra/badminton-highlight-indexer` (read the full diff). Also `gh pr view 420 -R Khelsutra/badminton-highlight-indexer --json body,files,statusCheckRollup` if useful.
2. Read the CURRENT master version of every file the PR touches (Read / Grep the files under C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer). To see the PR's branch version of a file without checkout, use: `git -C "C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer" show origin/docs/reconcile-trackers-2026-07-02:<path>`.
3. Determine whether master ALREADY implements the PR's intent (superseded) or still lacks it. Cite concrete master file:line evidence either way. Do not trust the hint or handoff ? verify in code.
4. If CONFLICTING, explain precisely why (which files/hunks) and whether the conflict is because master already did the same thing (=> superseded/close) or because of adjacent edits (=> rebasable).

5. Judge correctness/quality of the PR change itself.
6. Recommend a fate:
 - CLOSE_SUPERSEDED: master already implements the intent; the PR adds nothing material.
 - TAKE_OVER_CLEAN: mergeable & still-needed; adopt and verify as-is.
 - TAKE_OVER_REBASE: still-needed but conflicting; needs a rebase onto master + re-verify, then merge.
 - OWNER_DECISION: product/strategy/ambiguous ? needs the owner.

Return ONLY the structured object. Be precise and cite master file:line in master_evidence.

2. assistant (2026-07-02T08:16:27.606Z)

I'll investigate PR #420 to determine its fate. Let me start by getting the PR's intended change and understanding what files it touches.

3. user (2026-07-02T08:16:30.486Z)

<persisted-output>

Output too large (33.5KB). Full output saved to: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\tool-results\bkjsmmey7.txt

Preview (first 2KB):

```
diff --git a/backend/pipeline/segmenters/yolo_hybrid.py
b/backend/pipeline/segmenters/yolo_hybrid.py
index a41c83b..556a653 100644
--- a/backend/pipeline/segmenters/yolo_hybrid.py
+++ b/backend/pipeline/segmenters/yolo_hybrid.py
@@ -9,8 +9,10 @@
```

```
Stage 1 originally used the AGPL-licensed ultralytics YOLO. To keep the hosted
service free of copyleft obligations (see docs/MONETIZATION_AUDIT.md and ADR-005),
detection uses torchvision's permissively-licensed (BSD) detectors and the
-association/tracking that YOLO.track() used to bundle in is provided by
-supervision's MIT-licensed ByteTrack.
+association/tracking that YOLO.track() used to bundle in is provided by ByteTrack
+via the standalone Apache-2.0 ``trackers`` package (D13/P12 migration, #386 ?
+previously supervision's ``sv.ByteTrack``, removed upstream in supervision 0.30;
+see ``person_detector.make_tracker`` for the behaviour-pinning details).
```

The registry key stays "yolo_hybrid" for back-compat with existing configs and the DB `detector` tags, even though no YOLO is involved anymore.

```
diff --git a/docs/BACKLOG.md b/docs/BACKLOG.md
index 29ecd98..b02d4c7 100644
```

```
--- a/docs/BACKLOG.md
+++ b/docs/BACKLOG.md
@@ -5,7 +5,7 @@ Single tracker for work we consciously deferred, so nothing gets lost between se
> Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all of these
to **GitHub issues** instead (or in addition).
> Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.
```

```
-_Last updated: 2026-06-23._
```

```
+_Last updated: 2026-07-02 (reconciled the three 2026-06-10-sweep rows below against shipped
code: #370/#380/#386)._
```

```
---
```

```
@@ -19,8 +19,8 @@ _Last updated: 2026-06-23._
```

```
- ? **Define the ship-gate F1@tIoU target** ? "very high P/R" must be operationalized (floor =
beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). **Revisit:** at M0.4 harness.
```

```
## Engineering hardening deferred from the 2026-06-10 audit sweep
```

```
...
```

</persisted-output>

4. user (2026-07-02T08:16:32.124Z)

```
{"body":"## What / why\n\nDocs-only reconcile PR from the 2026-07-02 workspace audit (findings  
**R1-13**, **R1-14**, the **doc half of R1-01**, the **DATA_IN_GCS $5-vs-$7 contradiction**, and  
the dropped-pool `yolo_hybrid` docstring item). The session-start surfaces (DOC_STATUS / docs  
README / NEXT_STEPS / BACKLOG) had drifted ~15 merged PRs behind the per-doc source-of-truth  
trackers, which was already causing re-litigation of finished work (the project memory feeding  
the audit re-listed D13/ByteTrack as open). Every flip below was verified against code/git  
before writing; **no behavior change**.\n\n- **docs/DOC_STATUS.md** ? cleanup-audit row: Phase 1  
COMPLETE (#352?#371) + Phase 2 nearly done (#373?#386), remaining **P10 + P16+**;  
COMPUTE_DECOUPLED_SERVING row: the M5+ default-OFF batch is merged (M5 #286 · M6 #287 · M8 #288  
· M9-auth #290, M11 skeleton #336, M12 backend #338), next = M7 (? owner spend) + owner  
residuals; **new row** for `DECISION_SNAPSHOT_REGRESSION.md` (D1?D11 LOCKED 2026-06-25, merged  
#383, zero implementation ? was missing entirely, violating the register's own $0 cadence).\n- **docs/README.md:41-42** ? cleanup-audit blurb: no longer `DRAFT, owner-gated` (also: its  
tracker is $6, not $7; 35 findings, not 33); snapshot-doc blurb: no longer `decisions in $2  
need locking`.\n- **docs/NEXT_STEPS.md** ? `Where we are` rewritten to the post-06-21 shipped  
state (ADR-014 ratified = M0 sign-off; the `owner M0 design sign-off pending` claim  
contradicted the same file's item 1). The owner **LAUNCH NO-GO block (#404) is preserved  
verbatim** (confirmed unmodified in the diff ? context line only).\n- **docs/BACKLOG.md** ?  
line-39 ByteTrack row flipped ? DONE (#386) and its obsolete supervision `<0.30.0` cap sentence  
**deleted** (cap already lifted: `requirements.txt` now `supervision>=0.21.0`, `trackers @  
git+?@2.5.0`). Per the audit verifier the other two 2026-06-10-sweep rows are **compound and  
only half-shipped**, so they are annotated ? PARTIAL rather than flipped: app-factory shipped  
(#370) but no lifespan + the globals shim remains; Gemini GC shipped (#380) but WSL staged-  
video/frame/cache retention is still unscheduled (#301 open; #407 is a guard, not GC).\n- **docs/CODE_CLEANUP_AUDIT_2026-06-24.md** ? P10/D1 currency refresh: `main.py` re-measured at  
**2198 lines** (1448 at audit; sizes refreshed in $3b/$6/$8.1) and the step-1 extraction **PR  
#390 recorded as stale/CONFLICTING** (verified via `gh pr view 390`: mergeable=CONFLICTING,  
DIRTY) ? close + re-cut from `origin/master`. Header status + the $7? $6 tracking-anchor typo  
fixed; changelog entry added. (D13 was **already ? #386** in the $6 tracker ? the brief's `mark  
D13 ?` needed no change.)\n- **docs/DATA_IN_GCS.md** ? $5 gap table reconciled with its own  
$7/$8: trajectories/labels/manifest marked ? in-bucket since 2026-06-22 (they were still shown  
?), remaining gap = Phase-2 videos (1/15) + Phase-3 eval-on-GCS plumbing; the stale  
`C:\\?\\Annotation Setup\\Collect\\` local path corrected (folder moved under `khelsutra-  
guru`); $6 Phase-1 marked DONE.\n- **backend/pipeline/segmenters/yolo_hybrid.py** ? module  
**docstring only**: tracking is ByteTrack via the standalone Apache-2.0 `trackers` package  
(D13/P12, #386), no longer `supervision's MIT-licensed ByteTrack` (licensing-accounting text  
was wrong). Note: the brief pointed at `person_detector.py`, but its docstring was already  
migrated in #386 ? the stale text lived in `yolo_hybrid.py`, matching the audit's dropped-pool  
item.\n\n## Verification\n\nRan locally in the worktree (branched from `origin/master` @  
e410136):\n- `python -m tools.check_md_links` ? **clean** (the CI `link-check` job's exact  
command).\n- `ruff check backend/pipeline/segmenters/yolo_hybrid.py` ? all checks passed.\n- `mypy backend/pipeline/segmenters/yolo_hybrid.py` ? no issues.\n- `python -m pytest  
tests/test_yolo_hybrid.py -q` ? **8 passed** (docstring-only change; import + suite  
unaffected).\n- Fact checks behind each flip: `wc -l backend/main.py` = 2198; `requirements.txt`  
lines 18/29 (`supervision>=0.21.0`, `trackers @ ?@2.5.0`); `create_app` at `backend/main.py:93`  
with module-level `app` at :90 and no lifespan; `gemini.py::cleanup_orphaned_files` present;  
`backend/flywheel.py` (#336) + `backend/storage/gdrive_api.py` (#338) exist; `gh pr view 390` =  
OPEN/CONFLICTING/DIRTY; `git log` for #286/#287/#288/#290/#336/#338/#370/#380/#383/#386; the  
moved `khelsutra-guru\\Annotation Setup\\Trajectories\\` dir exists (old `Projects\\Annotation  
Setup` path does not).\n- Full suite + coverage floor: left to CI (docs-only + one docstring; no  
runtime paths touched).\n\nGenerated with [Claude Code](https://claude.com/claude-code)", "file  
s": [{"path": "backend/pipeline/segmenters/yolo_hybrid.py", "additions": 4, "deletions": 2, "changeType  
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"}], "mergeStateStatus": "DIRTY", "mergeable": "CONFLICTING", "statusCheckRollup": [{"__typename": "Che  
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02T04:58:09Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-  
highlight-indexer/actions/runs/28566508378/job/84694806957", "name": "Ruff lint", "startedAt": "2026  
-07-
```

```
02T04:57:59Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-02T04:58:05Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28566508378/job/84694806943", "name": "Leak scan (PII / attribution guard)", "startedAt": "2026-07-02T04:57:58Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-02T04:58:04Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28566508378/job/84694806950", "name": "Markdown link check (live docs)", "startedAt": "2026-07-02T04:57:58Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-02T04:58:29Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28566508378/job/84694806968", "name": "Mypy (lenient baseline)", "startedAt": "2026-07-02T04:57:58Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-02T04:58:20Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28566508378/job/84694806975", "name": "Syntax sweep + import smoke (no GPU stack)", "startedAt": "2026-07-02T04:57:58Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-02T05:00:40Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28566508378/job/84694806941", "name": "Full test suite (offline, mocked)", "startedAt": "2026-07-02T04:57:58Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-02T04:58:24Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28566508378/job/84694807058", "name": "Golden fixtures cross-OS (ubuntu-latest)", "startedAt": "2026-07-02T04:57:59Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-02T04:59:16Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28566508378/job/84694806994", "name": "Golden fixtures cross-OS (windows-latest)", "startedAt": "2026-07-02T04:57:59Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-02T04:58:24Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28566508378/job/84694807000", "name": "Golden fixtures cross-OS (macos-latest)", "startedAt": "2026-07-02T04:57:59Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-02T04:58:30Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28566508378/job/84694806976", "name": "Dependency license scan (no copyleft in runtime deps)", "startedAt": "2026-07-02T04:57:58Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-02T04:59:04Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28566508378/job/84694806981", "name": "Built wheel smoke (install + launch)", "startedAt": "2026-07-02T04:57:58Z", "status": "COMPLETED", "workflowName": "CI"}]}
```

5. assistant (2026-07-02T08:16:34.674Z)

I have the full PR metadata. Let me read the full diff and the current master versions of the key files.

6. user (2026-07-02T08:16:36.930Z)

```
1      diff --git a/backend/pipeline/segmenters/yolo_hybrid.py
b/backend/pipeline/segmenters/yolo_hybrid.py
2      index a41c83b..556a653 100644
3      --- a/backend/pipeline/segmenters/yolo_hybrid.py
4      +++ b/backend/pipeline/segmenters/yolo_hybrid.py
5      @@ -9,8 +9,10 @@
6      Stage 1 originally used the AGPL-licensed ultralytics YOLO. To keep the hosted
```

```

7     service free of copyleft obligations (see docs/MONETIZATION_AUDIT.md and ADR-005),
8     detection uses torchvision's permissively-licensed (BSD) detectors and the
9     -association/tracking that YOLO.track() used to bundle in is provided by
10    -supervision's MIT-licensed ByteTrack.
11    +association/tracking that YOLO.track() used to bundle in is provided by ByteTrack
12    +via the standalone Apache-2.0 ``trackers`` package (D13/P12 migration, #386 ?
13    +previously supervision's ``sv.ByteTrack``, removed upstream in supervision 0.30;
14    +see ``person_detector.make_tracker`` for the behaviour-pinning details).
15
16    The registry key stays "yolo_hybrid" for back-compat with existing configs and the
17    DB `detector` tags, even though no YOLO is involved anymore.
18    diff --git a/docs/BACKLOG.md b/docs/BACKLOG.md
19    index 29ecd98..b02d4c7 100644
20    --- a/docs/BACKLOG.md
21    +++ b/docs/BACKLOG.md
22    @@ -5,7 +5,7 @@ Single tracker for work we consciously deferred, so nothing gets lost
between se
23    > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all
of these to **GitHub issues** instead (or in addition).
24    > Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.
25
26    -_Last updated: 2026-06-23._
27    +_Last updated: 2026-07-02 (reconciled the three 2026-06-10-sweep rows below against
shipped code: #370/#380/#386)._
28
29    ---
30
31    @@ -19,8 +19,8 @@ _Last updated: 2026-06-23._
32    - ? **Define the ship-gate F1@tIoU target** ? "very high P/R" must be operationalized
(floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). **Revisit:** at M0.4
harness.
33
34    ## Engineering hardening deferred from the 2026-06-10 audit sweep
35    -- ? **App-factory / lifespan + lazy torch-cv2 imports** ? `uvicorn backend.main:app`
leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice
(#16). **Revisit:** when the async slice starts.
36    -- ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps,
`*_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy.
**Revisit:** before any volume/batch processing.
37    +- ? **App-factory / lifespan + lazy torch-cv2 imports** ? ? **PARTIAL (2026-07-02
reconcile).** **Shipped:** the app-factory landed as `create_app(config, db)` with state on
`app.state` (**#370**, cleanup-audit D2/P5), and torch/torchvision now import lazily
(`person_detector.py::lazy_load_model`; the LIGHT tier is torch-free per packaging P2e). `cv2`
stays top-level **by decision** (opencv-python is a hard dependency ? see the
`person_detector.py` module docstring). *Residual:* no lifespan ? the module-level `app` plus
the transitional `db_instance`/`global_config` globals shim remain, and `backend.main` is still
mypy-quarantined (see `CODE_CLEANUP_AUDIT_2026-06-24.md` §8 P5 note). **Revisit:** with the
D1/P10 `main.py` split.
38    +- ? **WSL temp/frame/cache + Gemini remote-file GC** ? ? **PARTIAL (2026-07-02
reconcile).** **Shipped:** the Gemini half ? per-call `_safe_delete` (#373) + a startup orphan
sweep (`backend/providers/gemini.py::cleanup_orphaned_files`, **#380**, cleanup-audit 01/P11).
*Residual:* staged videos, PNG frame dumps, and `*_wasbcache` still have **no retention policy**
? `tools/cleanup_caches.py` (#299) sweeps them on demand only, nothing schedules it (#301 open),
and the free-space guard (#407) rejects rather than reclaims. **Revisit:** before any
volume/batch processing.
39    - ? **shlex-quote WSL config values** ? **DONE 2026-06-11 (#97):**
`detectors/base.py::wsl_tilde_quote` quotes everything after a leading `~/`, so quoting and
tilde expansion coexist; applied to every `~`-capable arg in both runners;
`tests/test_wsl_tilde_quoting.py` pins the command shapes (no quoted tilde can ever reach
`_wsl_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the
default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations
(repo_dir/conda_env) if config ever becomes per-tenant.
40
41    ---
42    @@ -36,7 +36,7 @@ _Last updated: 2026-06-23._

```

```

43     ## Detector swap follow-ups
44     *(source: PR #8, ADR-005)*
45
46     -- ? **`supervision` ByteTrack deprecation** ? `sv.ByteTrack` is deprecated as of
supervision 0.28.0 and **removed in 0.30.0** ; `requirements.txt` is pinned `<0.30.0` as a
stopgap. Migrate to `ByteTrackTracker` from the separate `trackers` package before lifting the
cap. **Revisit:** before bumping supervision past 0.30.
47     +- ? **`supervision` ByteTrack deprecation** ? **DONE 2026-06-26 (#386, cleanup-audit
D13/P12):** migrated `sv.ByteTrack` ? `trackers.ByteTrackTracker`
(`backend/pipeline/detectors/person_detector.py::make_tracker`, constructor params pinned to
preserve `sv.ByteTrack` behaviour; `trackers @ git+?@2.5.0` in `requirements.txt` ? the PyPI
wheel is broken). The supervision cap is lifted (`requirements.txt` now `supervision>=0.21.0` ).
48     - ? **Real-video validation of the swap** ? the torchvision + ByteTrack Stage-1 is
unit-tested with mocks but **not yet validated on actual footage** for detection/tracking
quality. (ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video,
so expect tuning.) **Revisit:** when a real test clip + the pipeline are run together.
49     - ? **`yolo_hybrid` naming** ? the registry key, `yolo_` config keys, and `yolo-
hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. **Revisit:** if/when a clean break is worth the migration.
50
51     diff --git a/docs/CODE_CLEANUP_AUDIT_2026-06-24.md
b/docs/CODE_CLEANUP_AUDIT_2026-06-24.md
52     index 06f18bd..7c4f9dd 100644
53     --- a/docs/CODE_CLEANUP_AUDIT_2026-06-24.md
54     +++ b/docs/CODE_CLEANUP_AUDIT_2026-06-24.md
55     @@ -5,9 +5,9 @@ Follows the PROJECT_DOC_TEMPLATE.md conventions.
56
57     # Code Cleanup ? 2026-06-24 Deep Audit Remediation
58
59     -> **Status:** `PHASE 1 COMPLETE` ? 14 items shipped + D2 app-factory + test fixes .
**Owner:** Avi . **Created:** 2026-06-24 . **Last updated:** 2026-06-25 (DoD/line-count accuracy
fixes ? see Changelog)
60     +> **Status:** `PHASE 1 COMPLETE` + **Phase 2 nearly done** (P8/P9/P11?P15/P17 merged;
remaining = P10 + P16+) . **Owner:** Avi . **Created:** 2026-06-24 . **Last updated:**
2026-07-02 (P10 currency refresh ? see Changelog)
61     > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to
`docs/archives/past_projects/` when DONE)
62     -> **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`.
63     +> **Tracking anchors:** $6 progress tracker is the source of truth; indexed in
`docs/README.md`.
64     > **Relation to existing docs:** Sibling of `docs/CODE_AUDIT_AND_TEST_HARDENING.md`
(the 2026-06-10 hardening sweep).
65     >
66     > **Phase 1 summary:** All structural refactorings landed. `backend.api.job_worker`
lifted from mypy quarantine. Suite: **1463/1463 green**. `ruff`: all checks passed. `mypy`:
quarantined modules unchanged (ratchet held).
67     @@ -66,7 +66,7 @@ The goal is to fix these systematically, with **zero behavior change**
in Phase
68
69     | ID | Severity | Area | Issue | Phase |
70     |---|-----|-----|-----|-----|
71     -| **D1** | HIGH | `backend/main.py` | **Massive god-module** (~1450 lines). CLI, API,
worker orchestration, upload, debug-clip, trim, reingest all in one file. | 2 |
72     +| **D1** | HIGH | `backend/main.py` | **Massive god-module** (~1450 lines at audit
time; **2198 as of 2026-07-02** ? still growing). CLI, API, worker orchestration, upload, debug-
clip, trim, reingest all in one file. | 2 |
73     | **D2** | HIGH | `backend/main.py` + `backend/api/job_worker.py` | **Optional module
globals** (`db_instance`, `global_config`). NPE-prone; mypy-quarantined. | **1** |
74     | **D3** | HIGH | `backend/config/models.py` | **Dual config access.** Typed Pydantic
model + legacy `.get()` dict shim. `extra='allow'` silently keeps typo'd keys. | **1** |
75     | **D4** | MEDIUM | `backend/config/loader.py` | Shallow top-level merge: a
`config.json` section REPLACES the defaults dict for that section ? new fields missed on
upgrade. | 2 |
76     @@ -195,7 +195,7 @@ Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One

```

```

smal
77      | ? | ? | ? | ? | ? | ? |
78      | **P8** | **B2: Error-flagging in Gemini provider metrics** (Phase 2) | P7 | No | ? |
[#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |
79      | **P9** | **B1: Remove fabricated data from motion segmenter** ? random
server/receiver/shot_count/speed ? `None`; events dropped | P7 | No | ? |
[#385](https://github.com/Khelsutra/badminton-highlight-indexer/pull/385) |
80      -| **P10** | **D1: Split `main.py` god-module** (~1450 lines ? smaller modules) | P7 |
No | ? | ? |
81      +| **P10** | **D1: Split `main.py` god-module** (1448 lines at audit; **2198 as of
2026-07-02** ? smaller modules) | P7 | No | ? | step-1 extraction
[#390](https://github.com/Khelsutra/badminton-highlight-indexer/pull/390) went
**stale/CONFLICTING** (main.py grew ~52% under it, 2026-06-26) ? close + **re-cut fresh from
`origin/master`** |
82      | **P11** | **O1: Gemini remote-file GC** ? per-call `_safe_delete` (#373) + startup
orphan sweep scoped to our `UPLOAD_TAG`-tagged files, grace-gated, HTTP-timeout-bounded | P7 |
No | ? | [#380](https://github.com/Khelsutra/badminton-highlight-indexer/pull/380) |
83      | **P12** | **D13: ByteTrack deprecation** ? migrated `sv.ByteTrack` ?
`trackers.ByteTrackTracker` (git dep; PyPI wheel broken) | P7 | No | ? |
[#386](https://github.com/Khelsutra/badminton-highlight-indexer/pull/386) |
84      | **P13** | **D4: Deep config merge** ? section-level replace ? recursive merge | P7 |
No | ? | [#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |
85      @@ -244,7 +244,7 @@ All items below are **Claude-doable**, small-PR-friendly, and
unblocked. Pick up
86      |---|---|-----|-----|
87      | ~3~ | ~**B1 / P9**~ | ~Remove fabricated data from `motion.py`~ ? **? DONE
(#385).** Random server/receiver/shot_count/speed ? `None`; events dropped. |
88      | ~4~ | ~**D13 / P12**~ | ~ByteTrack deprecation migration~ ? **? DONE.**
`sv.ByteTrack` ? `trackers.ByteTrackTracker` (git dep; PyPI wheel broken ? filed Roboflow
issue). |
89      -| 5 | **D1 / P10** | Split `main.py` god-module | ~1450 lines. D2 (app-factory) was the
necessary precursor ? now done. Extract: CLI handlers ? `backend/cli/`, upload routes ? already
in `backend/api/uploads.py`, debug helpers ? `scripts/`. |
90      +| 5 | **D1 / P10** | Split `main.py` god-module | **2198 lines as of 2026-07-02** (1448
at audit ? every endpoint PR lands here, so it compounds). D2 (app-factory) was the necessary
precursor ? now done. ? The step-1 extraction (**#390**, `backend/orchestration.py`) is
stale/CONFLICTING ? close it and re-cut from `origin/master`. Extract: CLI handlers ?
`backend/cli/`, the newer `/api/source`/`/api/assets`/`/api/capabilities` routes ?
`backend/api/`, debug helpers ? `scripts/`. |
91
92      ### Later (deferred)
93
94      @@ -279,4 +279,5 @@ All items below are **Claude-doable**, small-PR-friendly, and
unblocked. Pick up
95      - `2026-06-25` ? **Accuracy fixes (docs-only).** (1) $8 DoD P5 said the module globals
were "replaced with `app.state`"; corrected ? `create_app()` *added* `app.state` and lifted
`job_worker` from quarantine, but `db_instance`/`global_config` remain as a transitional shim
and `backend.main` is still mypy-quarantined (`backend/main.py:264`, `mypy.ini`). (2) `main.py`
size "~1300" ? "~1450" (now 1448 lines) in $3b/$6/$8.1.
96      - `2026-06-25` ? **P17 done (B2 finish).** The 4 callers that dropped provider error
metrics (`trajectory_hybrid.py`, `yolo_hybrid.py`, `gemini_refine.refine_window`,
`gemini_refine.full_video_rallies`) now check `metrics["error"]` and log loudly instead of
silently treating a provider failure as "0 rallies" ? matching the pattern the `gemini`
segmenter already uses. (`yolo_hybrid` was caught during PR #378 review ? it bound `metrics` but
never read it.) Suite 1484/1484 green; ruff + mypy clean. Follow-up: migrate the hybrids'
`process_video` return type from bare `List[Dict]` to the ABC `Result[...]` contract (motion.py
is already migrated; hybrids suppressed with `type: ignore[override]` for now).
97      - `2026-06-25` ? **P9 done (B1 finish).** Removed fabricated `random` data from the
motion segmenter ? server/receiver/shot_count/avg_speed_kmh ? `None`, serve/smash/foul/winner
events dropped entirely (motion was the only `add_event` caller). PR #385 review caught a real
regression: `run_process_and_stitch.py` crashed on `None >= min_shot_count` (the `.get(?, 0)`
default no longer applied) ? fixed with a None-guard matching the API path. Suite 1492/1492
green; ruff + mypy clean.
98      +- `2026-07-02` ? **P10 currency refresh (docs-only, audit-reconcile).**
`backend/main.py` re-measured: **2198 lines** (was 1448 on 2026-06-25 ? 10 PRs landed on it

```

since, incl. #400/#405?#409/#412/#413); sizes refreshed in §3b/§6/§8.1. Recorded on the P10 row that the step-1 extraction PR **##390** (``backend/orchestration.py``, cut 2026-06-25) is now **stale + CONFLICTING** (verified via ``gh pr view 390``: mergeable=CONFLICTING, DIRTY) and should be closed + re-cut fresh from ``origin/master``. Header status/tracking-anchor also corrected (Phase-2 progress; the tracker is **§6**, not §7). No code change.

99 - `2026-06-26` ? **P12 CI fix (#386)**. The D13/P12 PR failed the ``full-suite`` lane with ``ModuleNotFoundError: No module named 'trackers'``. Two root causes: (1) the two new tests hard-imported the standalone ``trackers`` package, but it's a git-only dep (PyPI wheel is metadata-only ? `roboflow/trackers#474`) deliberately omitted from ``pyproject.toml``, and the lane installs via ``pip install -e .[dev]``; (2) the git ref ``@v2.5.0`` was wrong ? `roboflow/trackers` uses bare version tags (``2.5.0``), so pip failed with ``pathspec 'v2.5.0' did not match``. Fixes: added a ``pip install trackers @ git+?@2.5.0`` step to both ``ci.yml`` and ``nightly.yml`` full-suite lanes (same out-of-band pattern as torch); fixed the tag in ``requirements.txt``; and guarded the package-importing test with ``pytest.importorskip("trackers")`` (matching the repo's ``obsreport`` soft-dep convention). All 10 CI checks green.

100 diff --git a/docs/DATA_IN_GCS.md b/docs/DATA_IN_GCS.md

101 index 90cd99a..aa3e48c 100644

102 --- a/docs/DATA_IN_GCS.md

103 +++ b/docs/DATA_IN_GCS.md

104 @@ -7,7 +7,8 @@

105 > the storage layer). The deploy layer's bucket names now live in one place (``deploy/gcp/buckets.env``).

106

107 **Status:** LIVE / in-progress (migration not complete). **Owner-gated** decisions in §7.

108 -**Last inventoried:** 2026-06-22 against ``gs://khelsutra-rally-corpus``.

109 +**Last inventoried:** 2026-06-22 against ``gs://khelsutra-rally-corpus`` (§5 gap table reconciled

110 +to the post-Phase-1 state 2026-07-02 ? it had kept showing the pre-Phase-1 gaps, contradicting §7/§8).

111 **Goal (owner, 2026-06-22):** run **all** training + inference from the cloud with **all data in GCS**,

112 so the local Windows box (and the intermittently-mounted external drive) is **no longer** a blocker.

113

114 @@ -114,28 +115,33 @@ A bare/relative path resolves under a configured ``input_root``/``input_backend``

115

116 ## 5. The gap ? what's still local-only (why the box is still a blocker)

117

118 -All of this is **tiny** except the originals ? the metadata that blocks cloud eval is < 25 MB total:

119 +> **Reconciled 2026-07-02**. Phase-1 (all corpus **metadata**) landed **2026-06-22** ? see §7/§8; this

120 +> table used to show the pre-Phase-1 gaps and contradicted them. Post-Phase-1 the remaining gap is

121 +> **videos** (Phase 2: 1/15 canonical) and the **Phase-3 eval-on-GCS plumbing** (eval is still

122 +> local-FS-only code, so even bucket-resident metadata isn't consumed by eval yet ? §7b#2/#3).

123

124 | Data | Where it is now | Size | In bucket? | Needed for |

125 |---|---|---|---|---|

126 -| 20 WASB trajectory CSVs | ``C:\?\Annotation Setup\Collect\Trajectories\`` | **17.5 MB** |
| ? none | eval/litmus (cached). **(No `cached` train source exists ? train synths or detector-extracts.)** |

127 -| golden manifest | ``output/human_lovo_manifest.json`` (C:-absolute paths) | 4.5 KB | ?
| eval corpus definition |

128 -| original-6 labels | ``F:[Khelsutra]?Training\Golden Labelled\*.rallies.csv`` | ~12 KB
| ? | train + eval |

129 -| 7 new-clip labels | repo ``output/*.rallies.csv`` (gitignored) ****** F: archive (backed up 2026-06-22)** | ~12 KB | ? partial/inconsistent | train + eval |

130 -| source videos / proxies | F: archive + Google Drive golden folder | **100s of GB** |
? 3 only | infer + train (real WASB extraction) |

131 -


```

132     -> The corpus *metadata* (trajectories + labels + manifest, ~25 MB) is what actually
blocks GPU-free
133     -> cloud eval and the served Gate-A litmus ? and it's trivially small to migrate. The
**videos** are the
134     -> only large migration, and they already exist on **Google Drive** (the golden-corpus
folder), so they
135     -> can go **Drive ? bucket from a cloud box**, keeping the local machine out of the path
entirely.
136     +| WASB trajectory CSVs | bucket **** local working copy `C:\?\khehsutra-
guru\Annotation Setup\Trajectories\` *(the old `C:\?\Projects\Annotation Setup\Collect\` path
is gone ? the folder moved under `khehsutra-guru\`)* | **17.5 MB** | ? 15 canonical
`trajectories/<name>.csv` (2026-06-22, $7.2) | eval/litmus (cached). *(No `cached` train source
exists ? train synths or detector-extracts.)* |
137     +| golden manifest | `gs://?/manifests/golden.json` (gs:// paths); the **local** litmus
still reads `output/human_lovo_manifest.json` (C:-absolute) | 4.5 KB | ? (2026-06-22, $7.3) ?
but consumed by **no code yet** ($7b#3) | eval corpus definition |
138     +| original-6 labels | bucket **** F: archive | ~12 KB | ? among the 15 canonical
`labels/<date>_<name>.rallies.csv` (2026-06-22, $7.1) | train + eval |
139     +| 7 new-clip labels | bucket **** repo `output/*.rallies.csv` (gitignored) **** F:
archive | ~12 KB | ? ditto (naming reconciled; the 7 stale `_proxy`/bare dupes MD5-verified +
deleted) | train + eval |
140     +| source videos / proxies | F: archive + Google Drive golden folder | **100s of GB** |
? **1/15 canonical** (`mahadevpura_1` demo'd) + 2 pre-existing non-canonical objects pending
verify-dupe-then-delete ($6) ? **the remaining migration gap** | infer + train (real WASB
extraction) |
141     +
142     +> Post-Phase-1, the corpus *metadata* (trajectories + labels + manifest) is **in the
bucket**; what
143     +> still blocks GPU-free cloud eval is the **Phase-3 plumbing** ($6 ? the eval drivers
cannot fetch
144     +> `gs://` yet). The **videos** are the only large migration left (Phase 2), and they
already exist on
145     +> **Google Drive** (the golden-corpus folder), so they can go **Drive ? bucket from a
cloud box**,
146     +> keeping the local machine out of the path entirely.
147
148     ---
149
150     ## 6. Migration plan (close the gap)
151
152     -**Phase 1 ? corpus metadata (?25 MB, do first, removes the eval blocker).** Stage all
trajectories,
153     -all labels, and a **bucket-relative eval manifest** into `gs://khehsutra-rally-corpus`.
Pending the $7
154     -naming decision. Sketch:
155     +**Phase 1 ? corpus metadata (?25 MB, do first, removes the eval blocker). ? DONE
2026-06-22**
156     +(see $7 decisions + $8 tracker; the sketch below is kept as the historical plan). Stage
all
157     +trajectories, all labels, and a **bucket-relative eval manifest** into `gs://khehsutra-
rally-corpus`.
158     +Sketch:
159
160     ```bash
161     # trajectories (decision-free; absent today)
162     diff --git a/docs/DOC_STATUS.md b/docs/DOC_STATUS.md
163     index 7de53d4..c292f5f 100644
164     --- a/docs/DOC_STATUS.md
165     +++ b/docs/DOC_STATUS.md
166     @@ -1,6 +1,6 @@
167     # Documentation Status & Archival Register
168
169     -> **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-
diligence**. **Last full review:** 2026-06-21.
170     +> **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-

```

diligence**. **Last full review:** 2026-06-21 (row-level reconcile vs the per-doc trackers: 2026-07-02).

171 > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a milestone / delivered / archived), and re-review at each clean-slate.

172 > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update it as work finishes** to prove we're done, and (c) **never archive a foundational doc without honestly updating it first** (owner directive, 2026-06-21).

173

174 @@ -30,7 +30,7 @@ Before a foundational/tracked doc moves to `docs/archives/`:

175 ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's \$7 is the completeness source-of-truth)

176 | Doc | Status | Archive-ready? | Next action |

177 |---|---|---|---|

178 -| `COMPUTE\_DECOUPLED\_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M4 SHIPPED** (#279/#280/#282/#283); M5+ default-OFF code batch authorized (2026-06-21) | No | M5 truly-local ? M6 Cloud Run ? M8 ledger ? M9-auth ? M11 flywheel ? M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap) |

179 +| `COMPUTE\_DECOUPLED\_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M4 SHIPPED** (#279/#280/#282/#283); the M5+ default-OFF code batch **landed**?: M5 #286 · M6 #287 · M8 #288 · M9-auth #290 (all ? ? code merged default-OFF, owner residual open), M11 flywheel skeleton #336, M12 gdrive-api backend #338 | No | M7 cloud-serving e2e on L4 (? owner spend) + the owner residuals per \$7: counsel ToS/DPA, real prices, OAuth app, real-run within the D17 cap |

180 | `RALLY\_QUALITY\_RESEARCH.md` | Foundation, **tracked & ACTIVE** (\$7) | No | Drives the R-series; complete when \$7.1 single quality number is attributed |

181 | `R2\_EVAL\_METRIC\_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |

182 | `R1\_MILESTONE\_LAUNCH\_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the R-cluster) | Keep with R-series for narrative; archive when the track wraps |

183 @@ -44,7 +44,8 @@ Before a foundational/tracked doc moves to `docs/archives/`:

184 | `EXECUTION\_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud Run dispatch |

185 | `GOLDEN\_REGRESSION\_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval fixtures |

186 | **Data layer** ? `docs/data\_pipeline/` (`GOLDEN\_SET\_IMPLEMENTATION` · `GOLDEN\_SET\_PHASE1\_VIDEOS` · `GOLDEN\_VIDEOS` · `VIDEO\_RECORDING\_GUIDELINES` · `GOLDEN\_DATA\_SHARING`) | LIVING (data, **no-ML**) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data\_pipeline/README.md`. Roora *vendor* docs archived ? `archives/roora-vendor/`. |

187 -| `CODE\_CLEANUP\_AUDIT\_2026-06-24.md` | **IN PROGRESS** ? Phase 1: P0/P1/P4/P6a shipped (#352/#355/#354/#356). Quick wins (Q4/Q5/Q6/Q8) next. | No | Q4?Q5?Q6?Q8 ? P6b ? P2 (segmenters) ? P3 (extra=forbid) ? P5 (app-factory) ? P7 verify |

188 +| `CODE\_CLEANUP\_AUDIT\_2026-06-24.md` | **IN PROGRESS** ? **Phase 1 COMPLETE** (14 items, #352?#371) + **Phase 2 nearly done**?: P8/P9/P11?P15/P17 merged (#373?#386). Remaining: **P10** (D1 `main.py` split ? step-1 PR #390 went stale/CONFLICTING, to be re-cut) + **P16+** (Phase 3) | No | Re-cut the P10 split from `origin/master` (ideally after DECISION_SNAPSHOT_REGRESSION P1?P4, which proves such refactors quality-neutral) ? Phase 3 (P16+) |

189 +| `DECISION\_SNAPSHOT\_REGRESSION.md` | **DESIGN ? decisions D1?D11 LOCKED (owner, 2026-06-25)**; doc merged #383; implementation not started (\$7: P1?P7 ?, P8 ?) | No | P1 (canonical decision-snapshot schema) + P2 (factor pure `build\_stitch\_plan()` out of `compiler.py`) ? de-risks the P10 `main.py` re-cut |

190 | `DESKTOP\_APP\_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-feasibility spike; relates to truly-local profile |

191 | `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |

192 | `PERSONALIZATION\_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB v5 store landed) |

193 @@ -73,7 +74,7 @@ Before a foundational/tracked doc moves to `docs/archives/`:

194 | `CODE\_AUDIT\_AND\_TEST\_HARDENING.md` · `HARDENING\_LOOP\_HANDOFF.md` · `DEFERRED\_HARDENING\_PLAN.md` | COMPLETED ? **top-level pointer stubs** (archived copies exist) | **? confirm intentional stubs (per #165) vs finish-archiving (finding #4)** |

195

196 ### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

197 -| `PROJECT\_DOC\_TEMPLATE.md` · `NEXT\_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` ·

```

`EVALUATION.md` · `ABLATION_LAYER.md` · `SETUP_NEW_MACHINE.md` · ``CREATING_REELS.md`` (? end-
user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) ·
``CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`` (? operator runbook ? `gs://` videos ? reels on L4
Cloud Run; verified e2e 2026-06-23) · `TRACKNET_WSL_SETUP.md` · `README.md`. *(2026-06-21
relocation: the data-layer guides moved ? `data_pipeline/`; the field-ops + Roora docs ?
`archives/{field-pmf,roora-vendor}/`.)*
198 +`PROJECT_DOC_TEMPLATE.md` · `NEXT_STEPS.md` (? "Where we are" refreshed 2026-07-02) ·
`BACKLOG.md` · `EVALUATION.md` · `ABLATION_LAYER.md` · `SETUP_NEW_MACHINE.md` ·
``CREATING_REELS.md`` (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier;
added 2026-06-23) · ``CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`` (? operator runbook ? `gs://`
videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET_WSL_SETUP.md` ·
`README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data_pipeline/`; the field-
ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*
199
200 ### 3e. Already archived (`docs/archives/`)
201 `past_projects/DECOUPLED_COMPUTE/` (? 2026-06-21, #270) · `past_projects/low-regret-
moves/` · ``field-pmf/`` (field-ops + customer-feedback, 2026-06-21) · ``roora-vendor/``
(annotation-vendor engagement, 2026-06-21) · `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`
· `CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` ·
`COMMERCIAL_READINESS_REVIEW.md` · `MONETIZATION_AUDIT.md` · `TEST_RUN_SUMMARY.md` · `research/`
· `checkpoints/` · `quality-iterations/`.
202 diff --git a/docs/NEXT_STEPS.md b/docs/NEXT_STEPS.md
203 index 51847d4..ed9934b 100644
204 --- a/docs/NEXT_STEPS.md
205 +++ b/docs/NEXT_STEPS.md
206 @@ -1,9 +1,9 @@
207 -# Next steps (live) ? refreshed 2026-06-21
208 +# Next steps (live) ? "Where we are" refreshed 2026-07-02 (pending list last fully
reviewed 2026-06-21/22)
209
210 -> ``How to read:`` this top section is the ``current cross-track state + prioritized
pending list`` (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed
``owned-model & rally-quality track`` plan (2026-06-13) is ``preserved below, still valid`` ? it
is now ``one of several`` active tracks. Live blow-by-blow: memory `current-state.md`. Doc
lifecycle/hygiene: `docs/DOC_STATUS.md`.
211 +> ``How to read:`` this top section is the ``current cross-track state + prioritized
pending list`` ("Where we are" reconciled 2026-07-02 against the per-doc §-trackers + git; the
numbered pending list below it was last fully reviewed 2026-06-21/22 ? its own ? marks are
current). The detailed ``owned-model & rally-quality track`` plan (2026-06-13) is ``preserved
below, still valid`` ? it is now ``one of several`` active tracks. Live blow-by-blow: memory
`current-state.md`. Doc lifecycle/hygiene: `docs/DOC_STATUS.md`.
212
213 -## ? Where we are (2026-06-21)
214 -``DECOUPLED_COMPUTE delivered on GCP + archived`` (#270); ``compute = GCP only``
(ADR-013, DO dropped). ``COMPUTE_DECOUPLED_SERVING`` subproject spun up (#272, ``ADR-014``) ?
one pure core + local/cloud profiles; ``owner M0 design sign-off pending``. ``Per-video
workspace P0`` shipped (#264); ``doc-status register + archival due-diligence`` in place (#273).
The ``khelsutra site + per-rally UI`` ship in parallel (sibling track). The ``rally-quality /
owned-model`` track (below) is the core ML work, gated on ``growing the golden corpus``.
215 +## ? Where we are (2026-07-02)
216 +``DECOUPLED_COMPUTE delivered on GCP + archived`` (#270); ``compute = GCP only``
(ADR-013, DO dropped). ``COMPUTE_DECOUPLED_SERVING`` is deep in implementation** (#272;
``ADR-014 RATIFIED = owner M0 sign-off, 2026-06-21``): M1?M4 shipped (#279/#280/#282/#283) and
the default-OFF batch ``merged`` ? M5 #286 · M6 #287 · M8 #288 · M9-auth #290, plus the ``M11
flywheel skeleton`` (#336) and the ``M12 `gdrive-api://` backend`` (#338); the real gap is ``M7
cloud-serving e2e (? owner spend)`` + the owner residuals (item 1 below;
`COMPUTE_DECOUPLED_SERVING/README.md` §7 is the tracker). The ``packaging track`` is bundle-
ready and mid-flight (item 0b). The ``2026-06-24 cleanup audit`` is Phase-1 COMPLETE + Phase-2
nearly done (#352?#386); remaining: ``P10`` `main.py` god-module split (step-1 PR #390 went
stale/CONFLICTING ? re-cut it) + Phase-3 (`docs/CODE_CLEANUP_AUDIT_2026-06-24.md` §6). ``Per-
video workspace P0`` shipped (#264); ``doc-status register + archival due-diligence`` in place
(#273). The ``khelsutra site + per-rally UI`` ship in parallel (sibling track). The ``rally-
quality / owned-model`` track (below) is the core ML work, gated on ``growing the golden
corpus`` ? and on the launch decision recorded next:

```

```

217
218     > ??? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is
**not** cleared for an external/commercial launch; we stay in **development**. This is the
expected default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the
**ship-gate target (#172)** ? currently unset, needs corpus growth (and the R2 toLF1 ablation,
item 8); **(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean
licensing). P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev
PRs ? launching. Re-decide only when both unblockers clear.
219
220     diff --git a/docs/README.md b/docs/README.md
221     index edb3e82..7763468 100644
222     --- a/docs/README.md
223     +++ b/docs/README.md
224     @@ -38,8 +38,8 @@
225     Completed plans, old research, ADRs, and checkpoints live in [archives/](archives/).
226
227     - [CODE_AUDIT_AND_TEST_HARDENING.md](CODE_AUDIT_AND_TEST_HARDENING.md) ? **COMPLETED**
(2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing
/ contract / architectural audit, 626-test infrastructure, positive test plans for the 5
identified issues, multi-layer regression protocol (including ablation/experiment layer), and
pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`
for the final archived state. (Top-level is now pointer stub per review; full content in
archives/.)
228     -- [CODE_CLEANUP_AUDIT_2026-06-24.md](CODE_CLEANUP_AUDIT_2026-06-24.md) ? ??? Tracked
project (DRAFT, owner-gated):** the next-wave cleanup project ? 33 audit findings across bugs,
tech debt, operational risks, and quick wins. Phase 1 = pure refactorings (typed-config
migration, app-factory, segmenter validator fix); Phase 2+ = logic fixes. §7 progress tracker is
the source of truth; sibling of the completed hardening sweep above.
229     -- [DECISION_SNAPSHOT_REGRESSION.md](DECISION_SNAPSHOT_REGRESSION.md) ? ??? Tracked
project (DRAFT, owner-gated ? decisions in §2 need locking):** a deterministic **quality-neutral
proof** ? freeze + compare the pipeline's **decision output** (rally segments + stitch/concat
plan) off a cached trajectory, GPU-free, in CI. The behavioral proof that lets a *genuine* (non-
AST-identical) refactor be confidently called quality-neutral. **Never hashes the encoded reel**
(non-deterministic). Extends the existing characterization snapshot (`test_golden_fixtures.py`)
from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly_reelcount.py`.
§7 tracker is the source of truth.
230     +- [CODE_CLEANUP_AUDIT_2026-06-24.md](CODE_CLEANUP_AUDIT_2026-06-24.md) ? ??? Tracked
project (IN PROGRESS ? Phase 1 COMPLETE #352?#371; Phase 2 nearly done #373?#386):** the next-
wave cleanup project ? 35 audit findings across bugs, tech debt, operational risks, and quick
wins. Phase 1 (pure refactorings: typed-config migration, app-factory, segmenter validator fix)
shipped; Phase 2 (logic fixes: Gemini error contract, fabricated-data removal, Gemini remote-
file GC, ByteTrack migration) shipped except **P10** (the `main.py` god-module split ? step-1 PR
#390 stale/CONFLICTING, to be re-cut); Phase 3 (P16+) remains. §6 progress tracker is the source
of truth; sibling of the completed hardening sweep above.
231     +- [DECISION_SNAPSHOT_REGRESSION.md](DECISION_SNAPSHOT_REGRESSION.md) ? ??? Tracked
project (decisions D1?D11 LOCKED 2026-06-25, merged #383; implementation not started ? §7 P1?P7
?):** a deterministic **quality-neutral proof** ? freeze + compare the pipeline's **decision
output** (rally segments + stitch/concat plan) off a cached trajectory, GPU-free, in CI. The
behavioral proof that lets a *genuine* (non-AST-identical) refactor be confidently called
quality-neutral. **Never hashes the encoded reel** (non-deterministic). Extends the existing
characterization snapshot (`test_golden_fixtures.py`) from segmentation to the stitch plan; peer
of the M4 quality-floor + GPU `nightly_reelcount.py`. §7 tracker is the source of truth.
232
233     - **Field-ops / customer-feedback** (recruiting flyer · field-visit playbook · PMF) ?
[archives/field-pmf/`](archives/field-pmf/) (archived 2026-06-21; PMF validation still an
active *business* thread ? see `NEXT_STEPS.md`).
234     - **Roora annotation-vendor** engagement (labeling brief · vendor bake-off) ?
[archives/roora-vendor/`](archives/roora-vendor/) (archived 2026-06-21).
235

```

7. assistant (2026-07-02T08:16:41.658Z)

Now I have the full PR diff. This PR touches 7 files. Note that the PR body mentions #423 already registered the plan. The key question is whether master (which now includes #423)

already reflects these reconciliations. Let me read the current master versions of each file and compare with the PR's base (the PR was branched from `origin/master @ e410136`, but master is now at d00b4db which includes #423-#428).

Let me check the git history and current state of each file.

8. user (2026-07-02T08:16:45.102Z)

```
d00b4db fix gcs storage lazy import typing (#428)
19ebe5a clean remote input scratch dirs (#427)
c6bb4bf bound ffmpeg subprocess timeouts
6c767b5 cache remote source material
9523d17 fix process idempotency cache
e517475 docs: track khelsutra-guru health remediation plan
4f148d1 fix: config-correctness + typing fixes (audit R1-19, R1-20, R1-27) (#422)
d80aa02 docs: truth-sync ? audit report + fix stale state in CLAUDE.md, BACKLOG, DOC_STATUS,
README, NEXT_STEPS (#421)
e410136 api: /api/videos surfaces fps + resolution (from the stored ffmpeg details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)
aa915a7 deploy/gcp: centralize project + bucket names in one sourced buckets.env (#410)
52d97b5 P4: POST /api/assets ? store a video into the user's chosen medium (#409)
7ea6ea9 P3: /api/capabilities reports usable storage backends (#408)
f28af73 P2: engine free-space guard (HTTP 507 + WSL frame-stage BLOCK) (#407)
---MERGE BASE---
e4101365cb5f3a608e76f8baf1d2e0d65ce1738e
---PR BRANCH TIP---
dfdd2d9 docs: reconcile stale trackers/blurbs with shipped state (audit
R1-13/R1-14/R1-01-doc/DATA_IN_GCS)
e410136 api: /api/videos surfaces fps + resolution (from the stored ffmpeg details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
```

9. assistant (2026-07-02T08:16:50.493Z)

Key finding: The PR branched from e410136, which is BEFORE #421 (d80aa02 "docs: truth-sync") and #423 (e517475 "docs: track khelsutra-guru health remediation plan"). Both #421 and #423 touched these same docs. So the conflict is likely because master already did some of these reconciliations via #421/#423.

Let me examine the current master state of each file the PR touches. Let me start with the docs files and check the specific sections.

10. user (2026-07-02T08:16:52.649Z)

```
=== What did #421 (d80aa02) touch? ===
commit d80aa02e81c2c02e448ef75fa8777aa9c4561531
Author: Avi <avi.dullu@gmail.com>
Date: Thu Jul 2 10:55:33 2026 +0530
```

docs: truth-sync ? audit report + fix stale state in CLAUDE.md, BACKLOG, DOC_STATUS, README, NEXT_STEPS (#421)

```
Audit-driven doc-rot reconcile (AUDIT_REPORT R2-5, R1-13, R1-14):
- CLAUDE.md 'Current state': rewrite from 'No platform/owned-model implementation
  has started' to the shipped reality (async jobs, storage/compute backends,
  flywheel, Gen-0 weights, launch NO-GO 2026-06-30)
- CLAUDE.md: Gate A 'not wired' ? 'implemented in served_gate_a.py, default-OFF';
  StorageBackend/ComputeBackend 'Future' ? 'implemented'
- BACKLOG.md: ByteTrack deprecation ? DONE (#386); app-factory split (create_app
  shipped #370, lifespan+lazy-imports remain); Gemini GC shipped (#380), WSL
  temp/frame retention remains
- DOC_STATUS.md: CODE_CLEANUP_AUDIT row updated (Phase 1 COMPLETE, Phase 2 nearly
  done); added DECISION_SNAPSHOT_REGRESSION row (decisions locked #383);
  COMPUTE_DECOUPLED_SERVING row updated (M1-M9 largely shipped)
```

- README.md: CODE_CLEANUP_AUDIT + DECISION_SNAPSHOT blurbs updated to current state
- NEXT_STEPS.md: 'owner M0 design sign-off pending' ? 'M0-M9 LARGELY SHIPPED'
- Add the multi-repo audit report as a tracked project doc

```

CLAUDE.md | 30 +-
docs/AUDIT_REPORT_khelsutra-guru_2026-07-02.md | 605 +++++
docs/BACKLOG.md | 8 +-
docs/DOC_STATUS.md | 5 +-
docs/NEXT_STEPS.md | 2 +-
docs/README.md | 4 +-
6 files changed, 633 insertions(+), 21 deletions(-)

```

```

=== What did #423 (e517475) touch? ===
commit e5174752fb3d9a4f72bae153efd9451d88014e64
Author: Avi <avi.dullu@gmail.com>
Date: Thu Jul 2 12:09:23 2026 +0530

```

docs: track khelsutra-guru health remediation plan

Summary

- add a tracked five-repo health remediation plan extending the 2026-07-02 audit
- record post-sync completed items, pushback/reclassification, wave sequencing, coverage policy, and SPA/UI screencast requirements
- register the plan in README, DOC_STATUS, and NEXT_STEPS; mark the audit as a snapshot and update CODE_MAP for the #422 output-dir behavior

Verification

- python -m tools.check_md_links
- git diff --cached --check
- GitHub CI: all checks green, including Full test suite (offline, mocked)

```

docs/AUDIT_REPORT_khelsutra-guru_2026-07-02.md | 10 +-
docs/CODE_MAP.md | 2 +-
docs/DOC_STATUS.md | 3 +-
docs/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN.md | 214 +++++
docs/NEXT_STEPS.md | 1 +
docs/README.md | 3 +-
6 files changed, 226 insertions(+), 7 deletions(-)

```

11. assistant (2026-07-02T08:16:56.788Z)

This is critical. #421 (d80aa02) explicitly did an "audit-driven doc-rot reconcile (AUDIT_REPORT R2-5, R1-13, R1-14)" ? the SAME audit findings R1-13/R1-14 that PR #420 claims to address. So #421 substantially overlaps with #420's intent. Let me examine each file in detail to determine what's already done vs still unapplied.

Let me look at the current master state of each file section that #420 modifies.

12. user (2026-07-02T08:16:58.634Z)

```

=== yolo_hybrid.py docstring (master) ===
"""Two-stage rally segmenter: torchvision person detection + ByteTrack tracking
(Stage 1, local) feeding an AI provider for semantic boundaries (Stage 2).

```

Stage-1 detection now lives in `backend/pipeline/detectors/person\_detector.py` (`PersonDetector`, a reusable rally-cue producer ? fusion P1) so the SAME person cue can be consumed by the future fusion segmenter; this segmenter just orchestrates chunking, the AI-provider handoff, and DB population around it. Behaviour is unchanged.

Stage 1 originally used the AGPL-licensed ultralytics YOLO. To keep the hosted service free of copyleft obligations (see docs/MONETIZATION_AUDIT.md and ADR-005), detection uses torchvision's permissively-licensed (BSD) detectors and the association/tracking that YOLO.track() used to bundle in is provided by supervision's MIT-licensed ByteTrack.

The registry key stays "yolo_hybrid" for back-compat with existing configs and the DB `detector` tags, even though no YOLO is involved anymore.

""

```
import concurrent.futures
import logging
import os
import time
from typing import Any, Dict, List, Tuple

from backend.pipeline.detectors.person_detector import PersonDetector
```

13. assistant (2026-07-02T08:17:02.293Z)

The yolo_hybrid.py docstring on master STILL says "supervision's MIT-licensed ByteTrack" ? this hunk is NOT yet applied. #386 did NOT fix this docstring. So the yolo_hybrid.py change is still valid and unapplied.

Now let me check the docs files. Let me examine BACKLOG.md, DOC_STATUS.md, README.md, NEXT_STEPS.md on master to see what #421/#423 already did.

14. user (2026-07-02T08:17:03.826Z)

=== BACKLOG.md (master) - relevant sections ===

Backlog ? deferred items to revisit

Single tracker for work we consciously deferred, so nothing gets lost between sessions. Each item records **what**, **why** it's deferred, and **the trigger to revisit**. Every entry traces to a real decision/PR/doc (no speculative work).

> Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all of these to **GitHub issues** instead (or in addition).

> Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.

Last updated: 2026-07-02.

Batch / declarative orchestration

- ? **One-shot "serving startup-script"** for the GPU VM ? `create\_gpu\_vm.sh` already injects a metadata `startup-script` (`deploy/gcp/ops-agent-startup.sh`), but it only installs the **Ops Agent**; standing up the live engine on the VM (`kheksutra/deploy/GPU\_VM\_SERVING.md`) is still manual. Propose a `deploy/gcp/serving-startup.sh` fed via `--metadata-from-file=startup-script=?` (or pasted into the GCP console) that, on boot: installs ffmpeg/cv2 libs ? pulls the engine source (from GCS) + builds the venv ? **(optional)** the WASB native env (`gpu\_setup.sh` + weights) ? writes the serving `config.json`/`engine.env` ? installs `cloudflared` with the tunnel token ? starts `kheksutra-engine`. So a GPU serving box comes up **hands-off** (reuses the `deploy/gcp/nightly\_regression/run\_on\_vm.sh` bootstrap pattern). **Why deferred** ? the design hinge is **SECRETS**: the Gemini key / CF-Access AUD / tunnel token must **not** sit in a plaintext startup-script (instance metadata is broadly readable by anyone with `instances.get`); they should come from **Secret Manager** (VM SA ? `secretAccessor`) or be injected post-boot. Needs that small secrets-from-metadata design + which config is baked vs fetched ? not just a script (owner asked 2026-06-23). **Revisit**: when the GPU-VM serving path is used often enough to automate.

- ? **One YAML manifest drives the whole bucket?reels run** ? today, converting `gs://` videos into highlight reels is a sequence of API calls (see CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md). Propose a declarative **batch manifest** (`pipeline.yaml`): **inputs** (bucket prefix/glob or explicit list), **compute** target (local/cloud), **detection** (segmenter, skip_ai_handoff), per-video **selection** rules (all / top-N longest / min duration / a query string), and **output** (bucket prefix or local dir + watermark locale) ? driven by one `indexer --batch pipeline.yaml` runner that loops process?query?select?compile?deliver, **idempotent/resumable** (skips already-done `video\_id`s), reusing the existing API + the compute-picker + the compute-path provenance.

****Why deferred:**** needs a small design ? the manifest schema, the selection grammar, idempotency/resume, and partial-failure handling ? i.e. a tracked project doc per `PROJECT\_DOC\_TEMPLATE.md`, not a quick add (owner agreed 2026-06-23: ship the runbook doc now, backlog the YAML driver). ****Revisit:**** when hands-off batch conversion is wanted ? start with a design doc.

Owned-model / multisport (source: `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`, `NEXT\_STEPS.md`)

- ? ****Rename the repo off "badminton"*** ? owner-emphatic "ASAP" action; the project is multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. ****Revisit:**** owner action, before broad external sharing.
- ? ****Quantify cost-to-Gen-2 (the largest open risk)**** ? benchmark real VRAM/token/\$ for the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 6?8 GPUS) ****before**** standing up any owned serving stack. ****Revisit:**** before Phase-2 commitment.
- ? ****Define the ship-gate F1@tIoU target**** ? "very high P/R" must be operationalized (floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). ****Revisit:**** at M0.4 harness.

Engineering hardening deferred from the 2026-06-10 audit sweep

- ? ****App-factory + lazy torch-cv2 imports**** ? `create\_app()` shipped (#370), but `backend/main.py` still has a module-level app (no lifespan) and `person\_detector.py` still top-level-imports cv2. ****Revisit:**** when the async slice starts.
- ? ****WSL temp/frame/cache retention + Gemini remote-file GC**** ? Gemini orphaned-upload GC shipped (#380, `backend/providers/gemini.py::cleanup\_orphaned\_files` with a startup sweep). Remaining: staged videos, PNG frame dumps, and `\_wasbcache` still have no retention/cleanup. ****Revisit:**** before any volume/batch processing.
- ? ****shlex-quote WSL config values**** ? ****DONE 2026-06-11 (#97):**** `detectors/base.py::wsl\_tilde\_quote` quotes everything after a leading `~/`, so quoting and tilde expansion coexist; applied to every `~-`-capable arg in both runners; `tests/test\_wsl\_tilde\_quoting.py` pins the command shapes (no quoted tilde can ever reach `\_wsl\_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations (repo_dir/conda_env) if config ever becomes per-tenant.

Monetization / licensing

(source: `archives/MONETIZATION\_AUDIT.md` §7, ADR-005)

- ? ****Ultralytics Enterprise pricing**** ? get a quote ****only if**** the torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently ***not needed*** (swap done). Quote-only pricing. ****Revisit:**** if detector quality forces reconsidering YOLOv8.
- ? ****Codec patent thresholds (AVC/HEVC/AAC)**** ? confirm exact Via-LA (AVC) / Access Advance (HEVC) royalty rates & free thresholds ****with legal counsel**** before relying on the "small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus output. ****Revisit:**** before commercial launch.
- ? ****Per-checkpoint weight/dataset licenses**** ? if we borrow pretrained weights (YOLOX, ActionFormer, WASB, TrackNetV3), the ***code*** license ? the distributed ***weights/dataset*** terms. Either train our own or confirm each checkpoint. ****Revisit:**** when adopting any borrowed weights.
- ? ****Real GPU \$/video & latency**** ? benchmark the three LLM-step options (offline ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing the product. ****Revisit:**** when choosing the production LLM step.

Detector swap follow-ups

(source: PR #8, ADR-005)

- ? ****`supervision` ByteTrack deprecation**** ? ****DONE 2026-06-27 (#386):**** migrated to `ByteTrackerTracker` from the separate `trackers` package (`backend/pipeline/detectors/person\_detector.py`); the `<0.30.0` supervision cap is lifted (`supervision>=0.21.0`); `trackers @ git+...@2.5.0` pinned in `requirements.txt`.
- ? ****Real-video validation of the swap**** ? the torchvision + ByteTrack Stage-1 is unit-tested with mocks but ****not yet validated on actual footage**** for detection/tracking quality. (ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so expect tuning.) ****Revisit:**** when a real test clip + the pipeline are run together.

- ? **``yolo\_hybrid``** naming ? the registry key, **``yolo\_``** config keys, and **``yolo-hybrid-``** DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining. Optional rename. **Revisit:** if/when a clean break is worth the migration.

Golden-set subproject

*(source: **``docs/GOLDEN\_SET\_IMPLEMENTATION.md``**, **``docs/GOLDEN\_SET\_VENDORING.md``**)*

- ? **Annotation guideline** ? **DONE (2026-06-02):** authored as the Roora brief + illustrated guide in the collector repo **``docs/annotation/``** ? rally CSV schema; ignore dead time / include every rally (incl. faults & lets) / shot = one contact; **``ending\_reason``** (forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for badminton & pickleball.

- ? **Internal gold key** ? hand-annotate 5?10 licensed clips to near-perfect as the vendor answer key. **Unblocked.** **Revisit:** anytime.

- ? **GS-3** ? **``HumanVendorProvider``** ? concrete adapter for the chosen annotation vendor (submit/poll/fetch + DPA/secure-link controls). **Vendor** selected for the initial pilot: Roora (Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). **Revisit:** once the Roora pilot confirms quality, then build the adapter.

- ? **GS-4 oracle model** ? **``AutomatedProvider``** interface is stubbed (this work) with a deterministic fake. Wire a **concrete** fast-tier auto-labeler later. **Oracle rule:** it **MUST** be a different model lineage than production, or the regression test is blind to shared failures. **Revisit:** when a production LLM step is chosen (pick the oracle's lineage to differ).

- ? **GS-5** ? regression triggers ? wire **``regress()``** into CI (pre-merge gate), on-demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path. **Revisit:** after GS-4.

15. assistant (2026-07-02T08:17:11.722Z)

This is very telling. Master's BACKLOG.md (via #421) already reconciled the exact three rows that #420 addresses:

1. **App-factory row**: master already says **``create\_app()``** shipped (#370), but **``backend/main.py``** still has a module-level app (no lifespan)... ? #420 wants to add a more elaborate "PARTIAL" annotation, but the substance (create_app shipped, lifespan remains) is ALREADY reconciled.
2. **WSL/Gemini GC row**: master already says "Gemini orphaned-upload GC shipped (#380)... Remaining: staged videos... no retention/cleanup." ? same substance already there.
3. **ByteTrack deprecation row**: master already flipped to "DONE 2026-06-27 (#386)... supervision cap lifted (**``supervision>=0.21.0``**)..." ? #420 wants to flip this to ? DONE 2026-06-26. Already done.

So the BACKLOG.md reconciliation is SUPERSEDED by #421. The PR's version even differs in details (dates: #386 done 2026-06-26 in PR vs 2026-06-27 in master).

Now let me check DOC_STATUS.md, README.md, NEXT_STEPS.md on master.

16. user (2026-07-02T08:17:14.285Z)

=== DOC_STATUS.md (master) header + relevant rows ===

Documentation Status & Archival Register

> **Status** ? **LIVING** ? the single board for doc **lifecycle + archival due-diligence**. **Last full review:** 2026-07-02.

> **Update cadence**: whenever a doc changes lifecycle state (created / shipped a milestone / delivered / archived), and re-review at each clean-slate.

> **Why this exists**: so we can (a) **review completeness** at a glance, (b) **update it** as work finishes to prove we're done, and (c) **never archive** a foundational doc without honestly updating it first (owner directive, 2026-06-21).

0. How to use this

- **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention. §3 is the full register.

- **Marking something complete?** Update the doc's own **\$7 progress tracker** (the source of truth for **its** completeness), then flip its row here, then ? if terminal ? run the \$2 archival checklist.
- **This register does NOT duplicate completeness detail** ? it indexes lifecycle + archive-readiness and points to each tracked doc's \$7.

1. Lifecycle legend

``DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED``. Plus: **LIVING** (permanent reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal record of a run/launch), **CONVENTION** (a standing rule/guide).

2. Archival due-diligence checklist ? **the repeatable process (do NOT just `git mv`)**

Before a foundational/tracked doc moves to ``docs/archives/``:

1. **VERIFY** claims against the code ? no stale assertions; tag load-bearing facts ``[verified]``. (A doc that says "nothing built" when code shipped is **not** archive-ready ? fix it first.)
2. **Flip Status** ? **DELIVERED / DONE / REJECTED** + a completion note ? what shipped, where (PRs, paths, artifacts), and what was **deferred / carried forward**.
3. **Update ALL** inbound references ? ``docs/README.md`` index, ``CLAUDE.md``, ``docs/NEXT_STEPS.md``, and any cross-doc links ? the new path + status.
4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.
5. **`git mv`** the whole dir/file ? ``docs/archives/past_projects/`` (terminal-state **ONLY**; live docs stay at ``docs/`` top level). Relative links inside become historical (note this in the doc's header).
6. **Record the move** ? flip the row here to \$3e + a line in memory ``current-state``.

> **Worked example (the template):** **DECOUPLED_COMPUTE** ? #270 ? verified M0?M2+M3.5 delivered on GCP, flipped ??? **DELIVERED** with a completion note (incl. carried-forward WASB-C3/DPA/`md5`), reindexed ``docs/README`` + ``past_projects/README``, kept the ADR-010?014 lineage, then ``git mv`` to ``docs/archives/past_projects/DECOUPLED_COMPUTE/``.

3. The register

3a. Tracked projects ? lifecycle docs (archive when **DONE**; each doc's \$7 is the completeness source-of-truth)

Doc	Status	Archive-ready?	Next action
<code>`COMPUTE_DECOUPLED_SERVING/`</code>	IN PROGRESS ? M0 approved + M1?M9 LARGELY SHIPPED (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights pushed (<code>`weights/0.1.0-14/`</code>)	No	M11 flywheel + M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap)
<code>`RALLY_QUALITY_RESEARCH.md`</code>	Foundation, tracked & ACTIVE (\$7)	No	Drives the R-series; complete when \$7.1 single quality number is attributed
<code>`R2_EVAL_METRIC_DESIGN.md`</code>	IMPLEMENTED (augmenting)	No	(gates R4 + dual-eval; ablation pending) Promote-or-reject toIF1 as the gate (ablation) ? then archive with the R-cluster
<code>`R1_MILESTONE_LAUNCH_REPORT.md`</code>	LAUNCHED (#204) ? terminal record	Soon	(with the R-cluster) Keep with R-series for narrative; archive when the track wraps
<code>`RALLY_DETECTION_GAP_CLOSURE.md`</code>	DRAFT (owner-gated)	No	G1?G3 levers; part of the R/quality cluster
<code>`RALLY_DETECTION_FIXES_PLAN.md`</code>	DONE ? findings A/B/C resolved; P2b promoted <code>`inactivity_temporal_density`</code> default-ON (#396)	With the R-cluster (deferred per its header; gate = R2 toIF1 ablation + \$7.1 number)	Archive with the R-series when the track wraps
<code>`RALLY_DETECTION_CODE_REVIEW.md`</code>	Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (<code>`sdc#50`</code>) all resolved	With the R-cluster	Archive with the R-series
<code>`PER_VIDEO_WORKSPACE.md`</code>	DRAFT, IN PROGRESS (P0 = #264)	No	? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once)
<code>`MULTI_SIGNAL_FUSION_PLAN.md`</code>	P1 DONE; P2+ owner-gated	No	P3/P4 (learned fusion, audio) ? needs GPU golden features
<code>`EXPERIMENT_HARNESS.md`</code>	P1 IMPLEMENTED	No	A/B + promotion gate; supports the quality track
<code>`PROMINENT_COURT_DETECTION.md`</code>	DRAFT (owner-gated)	No	C-series; multi-track selection design

| `EXECUTION\_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud Run dispatch |
| `GOLDEN\_REGRESSION\_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval fixtures |
| **Data layer** ? `docs/data\_pipeline/` (`GOLDEN\_SET\_IMPLEMENTATION` ·
`GOLDEN\_SET\_PHASE1\_VIDEOS` · `GOLDEN\_VIDEOS` · `VIDEO\_RECORDING\_GUIDELINES` ·
`GOLDEN\_DATA\_SHARING`) | LIVING (data, **no-ML**) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data\_pipeline/README.md`. Roora *vendor* docs archived ?
`archives/roora-vendor/`. |
`CODE\_CLEANUP\_AUDIT\_2026-06-24.md`	**IN PROGRESS** ? Phase 1 COMPLETE (14 items shipped); Phase 2 nearly done (P8/P9/P11-P15/P17 merged #373-#386). Remaining: P10 (main.py split, stalled ? PR #390 CONFLICTING) + P16+ (Phase 3).	No	Re-cut P10 split; finish P16+ Phase 3 items
`KHELSUTRA\_GURU\_HEALTH\_REMEDIATION\_PLAN.md`	**IN PROGRESS** ? five-repo 2026-07-02 audit execution board; docs-only tracking bootstrap started; post-sync completions and pushback recorded	No	Execute waves 1-6 as one small PR per row; keep section 7 current
`DECISION\_SNAPSHOT\_REGRESSION.md`	**DRAFT ? decisions LOCKED** (D1?D11 locked 2026-06-25, #383). Zero code shipped for P1?P7.	No	Start P1+P2 (schema + stitch-plan factor); de-risk the P10 main.py split
`DESKTOP\_APP\_PLAN.md`	DRAFT (owner-gated, nothing built)	No	P0 compute-feasibility spike; relates to truly-local profile

17. assistant (2026-07-02T08:17:22.916Z)

DOC_STATUS.md on master is substantially reconciled by #421 and #423. Comparing to #420's intended changes:

- **Last full review date**: master says "2026-07-02" (via #421). #420 wants "2026-06-21 (row-level reconcile vs the per-doc trackers: 2026-07-02)". Different wording, but master is already updated to a 2026-07-02 review.
- **COMPUTE_DECOUPLED_SERVING row**: master already reconciled to "M1?M9 LARGELY SHIPPED (#279?#290...)". #420 wants a slightly different rewrite (adds M11 #336, M12 #338 explicitly in the status). Master's "Next action" mentions M11 flywheel + M12 gdrive-api but doesn't cite #336/#338. This is a minor detail difference ? the substantive reconciliation is done.
- **CODE_CLEANUP_AUDIT row**: master already says "Phase 1 COMPLETE... Phase 2 nearly done (P8/P9/P11-P15/P17 merged #373-#386). Remaining: P10... + P16+". Fully reconciled.
- **DECISION_SNAPSHOT_REGRESSION row**: master already ADDED this row (via #421): "DRAFT ? decisions LOCKED (D1?D11 locked 2026-06-25, #383)." #420 wanted to add this "new row" ? already present.

The key thing #420 claimed was a "new row for DECISION_SNAPSHOT_REGRESSION... was missing entirely" ? but master already has it (added by #421). So this is superseded.

Let me check the rest of DOC_STATUS (section 3d NEXT_STEPS note) and the README and NEXT_STEPS files.

18. user (2026-07-02T08:17:25.888Z)

=== DOC_STATUS.md section 3d (NEXT_STEPS refresh note) ===
21:3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`,
`docs/NEXT\_STEPS.md`, and any cross-doc links ? the new path + status.
78: `PROJECT\_DOC\_TEMPLATE.md` · `NEXT\_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` ·
`EVALUATION.md` · `ABLATION\_LAYER.md` · `SETUP\_NEW\_MACHINE.md` · **`CREATING\_REELS.md`** (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) ·
`CONVERT\_GCS\_VIDEOS\_TO\_HIGHLIGHTS.md` (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET\_WSL\_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data\_pipeline/`; the field-ops + Roora docs ?
`archives/{field-pmf,roora-vendor}/`.)*
90:2. **? RESOLVED (2026-06-21) ? `NEXT\_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.

=== README.md lines 36-45 (master) ===

Historical documents

Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).

- CODE_AUDIT_AND_TEST_HARDENING.md ? ****COMPLETED****
(2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing / contract / architectural audit, 626-test infrastructure, positive test plans for the 5 identified issues, multi-layer regression protocol (including ablation/experiment layer), and pre/post-change verification checklist. See `docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md` for the final archived state. (Top-level is now pointer stub per review; full content in archives/.)
- CODE_CLEANUP_AUDIT_2026-06-24.md ? ****?** Tracked project (Phase 1 COMPLETE, Phase 2 nearly done): ****** the next-wave cleanup project ? 33 audit findings across bugs, tech debt, operational risks, and quick wins. Phase 1 = pure refactorings (typed-config migration, app-factory, segmenter validator fix); Phase 2+ = logic fixes. \$7 progress tracker is the source of truth; sibling of the completed hardening sweep above.
- KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN.md ? ****?** Tracked project (IN PROGRESS, 2026-07-02): ****** live execution board for the five-repo `kheksutra-guru` audit cleanup. Extends the 2026-07-02 audit report, records post-sync completed items and pushback, then sequences the work into small PR waves with coverage and UI-screencast requirements. \$7 progress tracker is the source of truth.
- DECISION_SNAPSHOT_REGRESSION.md ? ****?** Tracked project (decisions D1?D11 LOCKED 2026-06-25, #383; implementation P1?P7 pending): ****** a deterministic ****quality-neutral proof**** ? freeze + compare the pipeline's ****decision output**** (rally segments + stitch/concat plan) off a cached trajectory, GPU-free, in CI. The behavioral proof that lets a ***genuine*** (non-AST-identical) refactor be confidently called quality-neutral. ****Never** hashes the encoded reel ****** (non-deterministic). Extends the existing characterization snapshot (`test\_golden\_fixtures.py`) from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly\_reelcount.py`. \$7 tracker is the source of truth.
- ****Field-ops / customer-feedback**** (recruiting flyer · field-visit playbook · PMF) ?
[`archives/field-pmf/`](archives/field-pmf/) (archived 2026-06-21; PMF validation still an active ***business*** thread ? see `NEXT\_STEPS.md`).

19. assistant (2026-07-02T08:17:34.632Z)

Now I can see the README.md situation more precisely. Comparing #420's intended README change to master:

#420 wanted:

- CODE_CLEANUP_AUDIT blurb: no longer "DRAFT, owner-gated"; tracker is \$6 not \$7; 35 findings not 33.
- DECISION_SNAPSHOT blurb: no longer "decisions in \$2 need locking".

Master (via #421) already:

- CODE_CLEANUP_AUDIT blurb: **"**?** Tracked project (Phase 1 COMPLETE, Phase 2 nearly done): ******" ? no longer "DRAFT, owner-gated". BUT still says "33 audit findings" (#420 wanted "35") and still says "\$7 progress tracker is the source of truth" (#420 wanted "\$6").
- DECISION_SNAPSHOT blurb: **"**?** Tracked project (decisions D1?D11 LOCKED 2026-06-25, #383; implementation P1?P7 pending): ******" ? no longer "decisions in \$2 need locking". Done.

So the README reconciliation is MOSTLY done by #421. Two small residual details in #420 are NOT in master: "35 findings" (vs 33) and "\$6 tracker" (vs \$7). But these are tiny fact-corrections, and the DRAFT?IN PROGRESS reconciliation (the bulk) is done.

Let me verify the \$6-vs-\$7 question and 33-vs-35 findings ? is the PR's claim even correct? Let me check the actual audit doc.

20. user (2026-07-02T08:17:39.014Z)

=== CODE_CLEANUP_AUDIT section headers (master) ===

```
17:## 0. TL;DR
23:## 1. Problem & goal
36:## 2. Decisions locked
50:## 3. Findings register (full audit)
123:## 4. Design / architecture ? Phase 1 approach
164:## 5. Honest limits ? what Phase 1 does NOT do
174:## 6. Deliverables & progress tracker ? **source of truth**
```

209:## 7. Open questions
216:## 8. Definition of done
230:## 8.1 ? Next items ? Phase 2 pick-up (prioritized)
259:## 9. References

=== findings count claims ===

109:These findings were already recorded in `docs/BACKLOG.md` as deferred items from the 2026-06-10 hardening sweep. They are absorbed into this project for execution:
261:- **Audit source:** 33 findings from a deep whole-repo read on 2026-06-24, plus 2 absorbed from `docs/BACKLOG.md` (D13, D14) for a total of 35. Two additional BACKLOG items (D2, 01) were already independently found and are cross-referenced in §3e.
273:- `2026-06-24` ? Absorbed 2 findings from `docs/BACKLOG.md` (D13, D14). Cross-referenced D2 and 01 in §3e.

21. assistant (2026-07-02T08:17:45.450Z)

This confirms two facts #420 asserted:

1. The progress tracker IS §6 ("## 6. Deliverables & progress tracker ? **source of truth**"), not §7. So master's README (via #421) STILL has the wrong "§7" reference ? #420's "§6" correction is still valid and unapplied.
2. The findings count IS 35 ("33 findings... plus 2 absorbed... for a total of 35"). So master's README "33 audit findings" is still slightly stale ? #420's "35" correction is still valid and unapplied.

So #420 has some residual value: the README §6/35 corrections, and the audit doc's own §7?§6 anchor fix + P10 currency refresh. Let me check the CODE_CLEANUP_AUDIT header/anchor state and the P10 rows on master, and NEXT_STEPS, and DATA_IN_GCS ? the two files #421/#423 touched less.

22. user (2026-07-02T08:17:48.384Z)

=== CODE_CLEANUP_AUDIT header (master lines 1-16) ===

```
<!--  
PROJECT DOC ? tracked code-cleanup + audit remediation plan.  
Follows the PROJECT_DOC_TEMPLATE.md conventions.  
-->
```

Code Cleanup ? 2026-06-24 Deep Audit Remediation

```
> Status: `PHASE 1 COMPLETE` ? 14 items shipped + D2 app-factory + test fixes . Owner: Avi . Created: 2026-06-24 . Last updated: 2026-06-25 (DoD/line-count accuracy fixes ? see Changelog)  
> Lifecycle: `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/past_projects/` when DONE)  
> Tracking anchors: §7 progress tracker is the source of truth; indexed in `docs/README.md`.  
> Relation to existing docs: Sibling of `docs/CODE_AUDIT_AND_TEST_HARDENING.md` (the 2026-06-10 hardening sweep).  
>  
> Phase 1 summary: All structural refactorings landed. `backend.api.job_worker` lifted from mypy quarantine. Suite: 1463/1463 green. `ruff`: all checks passed. `mypy`: quarantined modules unchanged (ratchet held).  
  
---
```

=== P10 / D1 rows and main.py size mentions (master) ===

```
10:> Tracking anchors: §7 progress tracker is the source of truth; indexed in `docs/README.md`.  
69:| D1 | HIGH | `backend/main.py` | Massive god-module (~1450 lines). CLI, API, worker orchestration, upload, debug-clip, trim, reingest all in one file. | 2 |  
198:| P10 | D1: Split `main.py` god-module (~1450 lines ? smaller modules) | P7 | No | ? | ? |  
221:- [x] P5: `create_app()` injects shared state via `app.state` (handlers read `request.app.state`); `backend.api.job_worker` removed from mypy quarantine. (The Optional module globals `db_instance`/`global_config` are retained as a transitional back-compat shim)
```

and `backend.main` stays mypy-quarantined ? full removal is follow-up, see D1/P10.)

247:| 5 | ****D1 / P10**** | Split `main.py` god-module | ~1450 lines. D2 (app-factory) was the necessary precursor ? now done. Extract: CLI handlers ? `backend/cli/`, upload routes ? already in `backend/api/uploads.py`, debug helpers ? `scripts/.` |

255:> ****Removed from this list ? already shipped (see the \$6 tracker); they were stale entries left from before Round 3/4 landed:**** D4/P13 deep config merge (#373 ? `loader.py::\_deep\_merge`), 02/P14 single-executor risk documented (#374 ? `job\_worker.py`), Q7/P15 `motion.py` type-cleaned (#373 ? no longer in the `mypy.ini` quarantine list).

278:- `2026-06-25` ? ****\$8.1 reconciled with the \$6 source-of-truth (docs-only).**** Removed three already-shipped items that \$8.1 still listed as deferred future work ? D4/P13, 02/P14, Q7/P15 (all merged in #373/#374) ? and split the unfinished half of B2 out of the done P8 into a new ****P17**** (the provider return contract + the `trajectory\_hybrid`/`gemini\_refine` callers that still drop `metrics["error"]`). Grounded against `loader.py::\_deep\_merge`, `mypy.ini`, `backend/results.py`, and `backend/providers/gemini.py` + its callers. No code change.

279:- `2026-06-25` ? ****Accuracy fixes (docs-only).**** (1) \$8 DoD P5 said the module globals were "replaced with `app.state`"; corrected ? `create\_app()` *added* `app.state` and lifted `job\_worker` from quarantine, but `db\_instance`/`global\_config` remain as a transitional shim and `backend.main` is still mypy-quarantined (`backend/main.py:264`, `mypy.ini`). (2) `main.py` size "~1300" ? "~1450" (now 1448 lines) in \$3b/\$6/\$8.1.

23. assistant (2026-07-02T08:18:00.349Z)

Important findings on CODE_CLEANUP_AUDIT_2026-06-24.md (master). Neither #421 nor #423 touched this file. So master's version is the ORIGINAL pre-#420 state:

- Header still says "Last updated: 2026-06-25" and "\$7 progress tracker is the source of truth" (line 10) ? #420 fixes both.
- P10 row (line 198) still shows "~1450 lines ? smaller modules" with "?" and "?" (no #390 stale note) ? #420 adds the 2198-line refresh + #390 stale note.
- D1 finding (line 69) still "~1450 lines" ? #420 refreshes to 2198.
- \$8.1 item 5 (line 247) still "~1450 lines" without #390 note.

Note: There's an internal inconsistency ? line 10 says "\$7" but the actual tracker is \$6 (and line 255/278 already reference "\$6 tracker"). So #420's \$7?\$6 fix on line 10 IS a valid correction.

So the CODE_CLEANUP_AUDIT_2026-06-24.md hunk of #420 is still ENTIRELY UNAPPLIED and valid. However ? is the "main.py = 2198 lines" claim in #420 still accurate on current master? #420 branched at e410136; master is now at d00b4db (with #422-#428 more merged). Let me verify current main.py size and #390 state.

24. user (2026-07-02T08:18:04.028Z)

```
=== current backend/main.py line count (master d00b4db) ===
2360 backend/main.py
```

```
=== #390 PR state ===
{"mergeStateStatus":"DIRTY","mergeable":"CONFLICTING","state":"OPEN","title":"refactor(D1/P10):
extract orchestration.py from main.py (god-module split, step 1)"}

```

```
=== requirements.txt supervision + trackers ===
18:supervision>=0.21.0
19:# D13/P12: ByteTrack moved out of `supervision` (removed in 0.30.0) to the standalone
20:# `trackers` package. The PyPI wheel is BROKEN (ships metadata only, no source ? see
21:# https://github.com/roboflow/trackers/issues/474), so install from git until fixed:
22:# pip install "trackers @ git+https://github.com/roboflow/trackers.git@2.5.0"
23:# (Apache-2.0; depends on supervision >=0.26.1, so it composes cleanly.)
24:# NOTE: roboflow/trackers uses bare version tags (2.5.0), NOT v-prefixed.
29:trackers @ git+https://github.com/roboflow/trackers.git@2.5.0

```

25. assistant (2026-07-02T08:18:09.927Z)

More important findings:

1. ****main.py is now 2360 lines on current master****, not 2198. So #420's "2198 as of 2026-07-02" claim is ALREADY STALE ? main.py grew further after the PR branched (#422-#428 landed more

code). The PR would re-introduce a number that's already out of date. This is a quality issue that argues against taking it over as-is.

2. ****#390 is still OPEN + CONFLICTING**** ? #420's statement that #390 is stale/CONFLICTING and should be re-cut is accurate.

3. ****requirements.txt**** confirms supervision>=0.21.0 and trackers @ git...@2.5.0 ? the PR's factual claims are correct.

Now let me check NEXT_STEPS.md and DATA_IN_GCS.md on master.

26. user (2026-07-02T08:18:11.446Z)

=== NEXT_STEPS.md (master, lines 1-20) ===

Next steps (live) ? refreshed 2026-06-21

> ****How to read:**** this top section is the ****current cross-track state + prioritized pending list**** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed ****owned-model & rally-quality track**** plan (2026-06-13) is ****preserved below, still valid**** ? it is now ****one of several* active tracks**. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene: `docs/DOC\_STATUS.md`.

? Where we are (2026-06-21)

****DECOUPLED_COMPUTE delivered on GCP + archived**** (#270); ****compute = GCP only**** (ADR-013, DO dropped). ****`COMPUTE\_DECOUPLED\_SERVING`**** subproject spun up (#272, ****ADR-014****) ? one pure core + local/cloud profiles; ****M0?M9 LARGELY SHIPPED**** (ADR-014 ratified 2026-06-21; M1?M4 #279?#283, M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF). ****Per-video workspace P0**** shipped (#264); ****doc-status register + archival due-diligence**** in place (#273). The ****khelsutra site + per-rally UI**** ship in parallel (sibling track). The ****rally-quality / owned-model**** track (below) is the core ML work, gated on ****growing the golden corpus****.

> ****? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).**** The rally-detection product is ****not**** cleared for an external/commercial launch; we stay in ****development****. This is the expected default for the dev phase. Two unblockers gate a future GO: **** (1) **** set + meet the ****ship-gate target (#172)**** ? currently unset, needs corpus growth (and the R2 tolF1 ablation, item 8); **** (2) **** resolve the ****WASB shuttle-detector weight provenance**** (commercial-clean licensing). P2b (#396) was a do-no-harm quality increment, ****not**** a launch gate ? merging dev PRs ? launching. Re-decide only when both unblockers clear.

? Prioritized pending ? all tracks

****P0 ? highest leverage / unblocks the most****

0. ****? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".**** Today the cloud flow is ****SSH/CLI-only**** (`docs/CLOUD\_GPU\_DRYRUN\_GUIDE.md`); ****M12 does NOT fix it.**** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **** (i) **** deploy the engine as a reachable ****service**** (M7 ? Cloud Run service, ? owner GCP spend for a ****persistent* endpoint**), **** (ii) **** the ****SPA ingest/progress screen**** (khelsutra `web/` B-P2: enter/upload video ? `api/process` ? poll `api/jobs` ? reel; ZERO engine code), **** (iii) **** flip ****M9 edge-auth ON**** (merged #290, default-OFF) to gate alpha tenants, **** (iv) **** a ****compute PICKER in the UI**** ("your machine vs our cloud", D13/D15). The engine seams make it ****possible****; the SPA makes it ****usable****. Full record: `[[ui-driven-cloud-serving-gap]]`.

0b. ****? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING\_PLAN.md`.**** The LIGHT-tier engine is bundle-ready: ``pip install`` ? ``indexer --app`` serves the ****bundled SPA**** single-origin, ****torch-free****, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). ****? Next (fresh focused pushes):**** **** (1) **** P3 ****GPU validation + screencast**** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), **** (2) **** P2b-install (uv/script + truly-local default config), **** (3) **** P2c first-run wizard (cross-repo khelsutra SPA from `api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

1. ? ****`COMPUTE\_DECOUPLED\_SERVING` M0?M9 LARGELY SHIPPED**** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), ****M5 #286, M6 #287, M8 #288, M9-auth #290**** all merged (default-OFF); the ****real M7 L4 run**** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). ****Owner residual**** (not blocking Claude's default-OFF code): counsel ToS/DPA

(M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **\$100/?10k cap** (D17).

2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.

3. **[owner + Claude · GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the `0.1.0-l4` bundle is n=2 plumbing-only).

P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)

3b. **[Claude] Five-repo audit health remediation** ? tracked in `docs/KHELSUTRA\_GURU\_HEALTH\_REMEDIATION\_PLAN.md` (created 2026-07-02). The audit report is a snapshot; the plan's §7 is the live source of truth. First wave = engine cost/correctness/resource fixes; later waves cover CI/supply-chain, SPA UX with screencast proof, and structural cleanup.

4. **[Claude] `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12** (M5/M6/M8/M9-auth ? merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-eval?goldenize plumbing reusing `workspace.py::for\_video` + `backend/eval/experiment.compare` + `backend/eval/promotion.py::promote\_if\_served\_safe`, behind `flywheel.\*`=OFF + a faked `consent\_attested` (? privacy-sensitive ? give it fresh focus). **M12** = `backend/storage/gdrive\_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex already allows the hyphen; add to `\_KNOWN` + `\_backend\_class` + the StorageConfig Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://` untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.

27. assistant (2026-07-02T08:18:19.862Z)

NEXT_STEPS.md on master (via #421) already reconciled the "Where we are" section:

- The stale claim "***owner M0 design sign-off pending***" is GONE ? master now says "***M0?M9 LARGELY SHIPPED**" (ADR-014 ratified 2026-06-21; M1?M4 #279?#283, M5 #286...).".
- Item 1 already says "M0?M9 LARGELY SHIPPED".
- The LAUNCH NO-GO block (#404) is preserved.
- Item 3b (via #423) added the five-repo health remediation reference.

#420 wanted the same reconciliation but with slightly different wording (adds M11 #336, M12 #338 explicitly, cites the cleanup audit Phase-1/2 state). The core stale-claim fix (removing "owner M0 design sign-off pending") is ALREADY done by #421. So NEXT_STEPS is substantially superseded, though the header line still says "refreshed 2026-06-21" (master) vs #420's "2026-07-02" ? a cosmetic date difference.

Now the last file: DATA_IN_GCS.md. Neither #421 nor #423 touched it. Let me verify.

28. user (2026-07-02T08:18:23.815Z)

=== git log touching DATA_IN_GCS.md (last 5) ===

8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)
db74f9b docs(data): mark DATA_IN_GCS §7b blocker #1 resolved by #319 (#321)
e221cb9 docs(data): GCS corpus source-of-truth map + Phase-1 metadata migration (#316)

=== DATA_IN_GCS.md master lines 5-12 + section 5 (114-160) ===

> `GET /api/capabilities.storage` (see `khehsutra/docs/STUDIO\_MEDIA\_LIFECYCLE.md`) ? and must stay physically
> isolated from the corpus bucket (the **D1** compliance boundary: consent / no-minors / not-training holds at
> the storage layer). The deploy layer's bucket names now live in one place
(`deploy/gcp/buckets.env`).

Status: LIVE / in-progress (migration not complete). **Owner-gated** decisions in §7.

Last inventoried: 2026-06-22 against `gs://khehsutra-rally-corpus`.

Goal (owner, 2026-06-22): run **all** training + inference from the cloud with **all data in GCS**,

so the local Windows box (and the intermittently-mounted external drive) is **no longer a blocker**.

---SECTION 5---

5. The gap ? what's still local-only (why the box is still a blocker)

All of this is **tiny** except the originals ? the metadata that blocks cloud eval is < 25 MB total:

Data	Where it is now	Size	In bucket?	Needed for
20 WASB trajectory CSVs	`C:\?\Annotation Setup\Collect\Trajectories\`	**17.5 MB**	? none	eval/litmus (cached). *(No `cached` train source exists ? train synths or detector-extracts.)*
golden manifest	`output/human_lovo_manifest.json` (C:-absolute paths)	4.5 KB	? eval	corpus definition
original-6 labels	`F:\[Khelsutra]\Training\Golden Labelled\*.rallies.csv`	~12 KB	? train + eval	
7 new-clip labels	repo `output/*.rallies.csv` (gitignored) + F: archive (backed up 2026-06-22)	~12 KB	? partial/inconsistent	train + eval
source videos / proxies	F: archive + Google Drive golden folder	**100s of GB**	? 3 only	infer + train (real WASB extraction)

> The corpus **metadata** (trajectories + labels + manifest, ~25 MB) is what actually blocks GPU-free
> cloud eval and the served Gate-A litmus ? and it's trivially small to migrate. The **videos** are the
> only large migration, and they already exist on **Google Drive** (the golden-corpus folder), so they
> can go **Drive ? bucket** from a cloud box, keeping the local machine out of the path entirely.

6. Migration plan (close the gap)

Phase 1 ? corpus metadata (~25 MB, do first, removes the eval blocker). Stage all trajectories, all labels, and a **bucket-relative eval manifest** into `gs://khelsutra-rally-corpus`. Pending the §7 naming decision. Sketch:

```
```bash
```

### trajectories (decision-free; absent today)

```
gcloud storage cp "C:\?\Collect\Trajectories*.csv" gs://khelsutra-rally-corpus/trajectories/
```

**labels (after the §7.1 naming decision) -> labels/<canonical-name>.rallies.csv**

**eval manifest with gs:// (or bucket-relative) paths -> a stable manifest object (§7.3)**

```
```
```

Phase 2 ? videos ? `videos/<name>.mp4` (bucket name = the eval `name`, consistent with `trajectories/` + `labels/`). Sources (resolved 2026-06-22 ? see `scratch/copy\_videos\_to\_gcs.ps1`):

| Source | Clips | Size |
|--|---|--------------|
| Drive golden folder | `~/Golden Label Videos Request - June 15/[Done] Video N/` (proxies) | |
| | `mahadevpura_1/2`, `GX010128`, `GX020094`, `GX030094`, `Badminton_BXH_2`, `Boxhill_Doubles` (7) | ~13.5 GB |
| F: archive (full MP4s ? proxy-first before upload) | `GX010137`, `GX010141`, + original-6 (`adarsh_avi_singles`, `kushagra_singles`, `largetest_doubles`, `gbaaddy`, `testlarge_short`, `mahadevpura_singles`) | (8) ~50 GB |

Two ways to run it:

- **Decoupled** (preferred ? no local box): ``rclone`` on a cloud VM with a Google-Drive remote ? ``rclone copy gdrive:"<?/[Done] Video N/><proxy>.mp4" :gs:khelsutra-rally-corpus/videos/<name>.mp4``

(one-time owner step: `rclone config` ? `gdrive` remote OAuth). Or `gdown <file-id>` on the box for

the shared folder. This is the path that actually removes the local box.

- **Interim (today, owner box ? streams Drive/F: ? local ? GCS):** `pwsh scratch/copy\_videos\_to\_gcs.ps1`

29. assistant (2026-07-02T08:18:29.935Z)

DATA_IN_GCS.md on master is the ORIGINAL pre-#420 state. Section 5's gap table STILL shows the pre-Phase-1 gaps:

- trajectory CSVs: "? none"
- golden manifest: "?"
- original-6 labels: "?"
- 7 new-clip labels: "? partial/inconsistent"
- source videos: "? 3 only"

And the stale local path `C:\?\Annotation Setup\Collect\Trajectories\` is still there. And §6 Phase 1 does NOT say "? DONE 2026-06-22".

So #420's DATA_IN_GCS.md hunk is entirely UNAPPLIED. But is #420's claim (metadata already in-bucket since 2026-06-22) CORRECT? The git log shows #316 "GCS corpus source-of-truth map + Phase-1 metadata migration" and #321 "mark §7b blocker #1 resolved". Let me verify §7/§8 of DATA_IN_GCS to confirm the internal contradiction #420 claims to fix is real.

30. user (2026-07-02T08:18:33.825Z)

=== DATA_IN_GCS §7 and §8 (search for in-bucket / DONE / 2026-06-22 evidence) ===

10:**Last inventoried.** 2026-06-22 against `gs://khehsutra-rally-corpus`.

11:**Goal (owner, 2026-06-22):** run **all** training + inference from the cloud with **all data in GCS**,

49:| `labels/` | golden rally labels (`<clip>.rallies.csv`) | **input** ground truth to `worker train` + all eval |

50:| `trajectories/` | cached WASB trajectory CSVs | ? **eval/litmus input** (GPU-free), consumed via `manifests/golden.json`. **NOT** read by `worker train` (it synths or re-extracts) |

51:| `weights/<version>/` | trained Gen-0 bundles (`weights.json` + a run `manifest.json`) | **output** of `worker train` |

56:> **Note on the manifest.** `worker train` does **not** read a pre-staged corpus `manifest.json` ? it

57:> *builds* one on-box at train time from `labels/` (+`videos/`), extracting trajectories **synthetically

59:> detector` (needs `videos/`). A `manifest.json` only appears as an **output** inside each

60:> `weights/<version>/` bundle. The **eval/litmus** path is what wants a stable, path-resolvable manifest

61:> (§7.3) ? and `manifests/golden.json` is currently consumed by **no code yet** (eval is local-FS only;

66:## 3. What's actually in the bucket today (2026-06-22)

68:> **Phase-1 landed 2026-06-22:** added `trajectories/` (15), 15 canonical `labels/<date>\_<name>.rallies.csv`

70:> `manifests/golden.json`. The table below is the **pre-Phase-1** snapshot; the remaining gap is **videos**

75:| `labels/` | 7 files: `GX010128`, `mahadevpura-1`, + `Badminton BXH 2\_proxy`, `Boxhill Doubles\_proxy`, `GX020094\_proxy`, `GX030094\_proxy`, `mahadevpura-2\_proxy` | ? **inconsistent naming** (`\_proxy` on some, not others); ? **missing** the original-6 labels + `GX010137`, `GX010141` |

77:| `trajectories/` | ? | ? **does not exist** (every trajectory is local-only) |

78:| `models/` | `wasb\_badminton\_best.pth.tar` | ? (? WASB-C3 weight-provenance still an open legal item before any external share) |

79:| `weights/` | `0.1.0-l4/` | ? first real L4 train bundle |

80:| `outputs/` | one `<video\_id>/`, `parity/` | ? dry-run outputs |

81:| `highlights/`, `promo/` | reels + promo | ? |

84: `weights/`) and **inconsistent `labels/` naming**, and is **missing the bulk of the corpus inputs**

85:(`videos/`, all `trajectories/`, most `labels/`). It reflects past *dry runs*, not a complete corpus.

```

103:# TRAIN from the bucket (reads labels/ [+ videos/], extracts trajectories, builds manifest
on-box,
122:| golden manifest | `output/human_lovo_manifest.json` (C:-absolute paths) | 4.5 KB | ? |
eval corpus definition |
124:| 7 new-clip labels | repo `output/*.rallies.csv` (gitignored) **** F: archive (backed up
2026-06-22) | ~12 KB | ? partial/inconsistent | train + eval |
127:> The corpus *metadata* (trajectories + labels + manifest, ~25 MB) is what actually blocks
GPU-free
137:all labels, and a **bucket-relative eval manifest** into `gs://khelsutra-rally-corpus`.
Pending the $7
142:gcloud storage cp "C:\?\Collect\Trajectories\*.csv" gs://khelsutra-rally-
corpus/trajectories/
143:# labels (after the $7.1 naming decision) -> labels/<canonical-name>.rallies.csv
144:# eval manifest with gs:// (or bucket-relative) paths -> a stable manifest object ($7.3)
148:`trajectories/` + `labels/`). Sources (resolved 2026-06-22 ? see
`scratch/copy_videos_to_gcs.ps1`):
165:? Two pre-existing non-canonical video objects (`videos/GX030094_proxy.mp4`,
`videos/mahadevpura-1.MP4`)
172:have **no storage-layer fetch**, so a `gs://` manifest cannot be consumed today. Phase-3
must thread
182:1. **`labels/` naming ? LOCKED (owner, 2026-06-22):** `labels/<YYYY-MM-
DD>_<name>.rallies.csv`, where
183:  `<name>` is the eval/manifest clip name and `<YYYY-MM-DD>` is the label's **authoring
date** =
188:2. **Cache `trajectories/` ? YES, done 2026-06-22.** `trajectories/<name>.csv` cached so
cloud/CI eval +
190:3. **Eval manifest in GCS ? done 2026-06-22.** `gs://khelsutra-rally-
corpus/manifests/golden.json`
191: (15 clips, `gs://` trajectory + label paths). The **local** litmus keeps its local-path
manifest;
192:  flipping the litmus to read the GCS manifest is Phase 3.
197:> authoring date, build the `gs://` manifest) ? `gcloud storage cp` (planner kept in
gitignored
199:> ? **Cleanup DONE 2026-06-22:** the 7 pre-existing inconsistent label objects
(`*_proxy.rallies.csv`,

```

31. assistant (2026-07-02T08:18:42.608Z)

This confirms #420's premise for DATA_IN_GCS.md is factually CORRECT and creates a real internal contradiction:

- \$3 (line 68) already notes "Phase-1 landed 2026-06-22: added trajectories/ (15), 15 canonical labels/..., manifests/golden.json. The table below is the pre-Phase-1 snapshot."
- \$7 (lines 188-192) confirms trajectories cached (done 2026-06-22), eval manifest in GCS (done 2026-06-22), labels naming locked.
- But \$5's gap table (lines 118-125) STILL shows "? none" / "?" for trajectories/manifest/labels ? contradicting \$3/\$7/\$8.

So #420's DATA_IN_GCS.md hunk fixes a REAL, still-present contradiction and is entirely unapplied. This is the most-valuable, non-superseded part of #420. Note: #420 says videos "1/15 canonical", but \$3/\$6 on master (lines 79-80, 165) reference dry-run outputs and "Two pre-existing non-canonical video objects" ? the exact video count is a nuance, but the metadata ??? correction is clearly valid.

Now I have enough to fully assess. Let me do a final confirmation of the conflict nature ? which hunks conflict with master's #421/#423 edits. Let me try a dry-run merge check to see exactly which files conflict.

32. user (2026-07-02T08:18:49.781Z)

=== per-file: does master (HEAD) differ from PR merge-base (e410136) for each touched file? (=> conflict candidates) ===

```

docs/BACKLOG.md : master-vs-base = 4+/4-
docs/DOC_STATUS.md : master-vs-base = 5+/3-
docs/README.md : master-vs-base = 4+/3-
docs/NEXT_STEPS.md : master-vs-base = 2+/1-

```

docs/CODE_CLEANUP_AUDIT_2026-06-24.md : master-vs-base = UNCHANGED
docs/DATA_IN_GCS.md : master-vs-base = UNCHANGED
backend/pipeline/segmenters/yolo_hybrid.py : master-vs-base = UNCHANGED

33. assistant (2026-07-02T08:19:12.937Z)

This crisply confirms the conflict analysis. The 4 files that master changed since the PR base (BACKLOG.md, DOC_STATUS.md, README.md, NEXT_STEPS.md) are exactly where #421/#423 already reconciled the SAME lines #420 edits ? these are the CONFLICTING hunks, and the conflict is because master already did the same reconciliation (superseded), not adjacent edits.

The 3 files master did NOT touch (CODE_CLEANUP_AUDIT_2026-06-24.md, DATA_IN_GCS.md, yolo_hybrid.py) are where #420 still has UNAPPLIED, non-superseded value ? those hunks would apply cleanly.

Let me do one final check to confirm the yolo_hybrid, DATA_IN_GCS, and CODE_CLEANUP hunks would actually apply on current master (context intact), by checking git can locate the diff context. I already confirmed the exact master content matches the diff's "-" lines for all three. Let me verify the CI check status meaning and finalize.

I have all the evidence needed. Let me compose the structured verdict.

Summary of findings:

****Superseded hunks (4 files, all CONFLICTING ? because #421/#423 already did the same reconciliation):****
- `docs/BACKLOG.md` ? #421 already flipped ByteTrack row to ? DONE (#386, cap lifted to `supervision>=0.21.0`), and already reconciled the app-factory row (create_app #370 shipped, lifespan remains) and WSL/Gemini-GC row (#380 shipped, retention remains). #420's versions are more verbose "? PARTIAL" annotations of the same facts. Superseded.
- `docs/DOC\_STATUS.md` ? #421 already added the DECISION_SNAPSHOT_REGRESSION row (which #420 called "missing entirely"), reconciled the CODE_CLEANUP_AUDIT row (Phase 1 COMPLETE / Phase 2 nearly done) and COMPUTE_DECOUPLED_SERVING row (M1?M9 shipped), and bumped "Last full review" to 2026-07-02. Superseded.
- `docs/README.md` ? #421 already flipped both blurbs off "DRAFT, owner-gated" / "decisions in §2 need locking". Two tiny residual fact-corrections in #420 remain unapplied (still says "33 findings" not 35, "§7" not §6 ? both verified against the audit doc), but the substantive reconciliation is done.
- `docs/NEXT\_STEPS.md` ? #421 already removed the stale "owner M0 design sign-off pending" and rewrote "Where we are" to M0?M9 shipped; #423 added the health-remediation item. Superseded (only a cosmetic header-date difference remains).

****Still-valid, unapplied hunks (3 files master never touched):****
- `backend/pipeline/segmenters/yolo\_hybrid.py` ? master docstring STILL says "supervision's MIT-licensed ByteTrack" (backend/pipeline/segmenters/yolo_hybrid.py:12-13). #386 did NOT fix this. #420's correction (Apache-2.0 `trackers` package) is accurate and unapplied.
- `docs/DATA\_IN\_GCS.md` ? §5 gap table (lines 118-125) STILL shows ? for trajectories/manifest/labels, directly contradicting §3 (line 68), §7 (lines 188-192), §8 which say Phase-1 metadata landed in-bucket 2026-06-22. Real, still-present contradiction; #420 fixes it. Unapplied and correct.
- `docs/CODE\_CLEANUP\_AUDIT\_2026-06-24.md` ? master still says "§7 progress tracker is the source of truth" (line 10) though the tracker is actually §6; still shows P10 as "~1450 lines / ? / ?" with no #390 stale note. #420 fixes the §7?§6 anchor and records #390 stale/CONFLICTING. Unapplied.

****Correctness/quality issue with the still-valid part:**** #420's main.py line-count claim ("2198 lines as of 2026-07-02") is ALREADY STALE ? current master `backend/main.py` is 2360 lines (grew via #422-#428 after the PR branched at e410136). Adopting as-is would re-introduce a wrong number.

Given the conflict is genuinely superseded on the 4 changed files, but 3 files carry real, still-unapplied fixes (with one stale number to correct), this needs a rebase that drops the superseded hunks and keeps/refreshes the still-valid ones.

34. user (2026-07-02T08:19:35.425Z)

Structured output provided successfully